

Optimal Impulsive Rendezvous Under Target Flyby Constraint Based on Primer Vector Theory

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This article investigates the two-body fuel-optimal impulsive rendezvous problem with a target flyby constraint. First, by defining the corresponding impulsive trajectory parameters, the constraint is implicitly incorporated into the problem formulation, and the feasibility of the constrained impulsive trajectory is ensured through solving the Lambert two-point boundary value problem. Second, to solve for the optimal impulsive trajectory, the constrained cost gradient and impulse addition criteria are derived based on the calculus of variations. In this derivation process, the constrained primer vector is defined, with the conventional primer vector proven to be a special form of the former in the unconstrained case. As a key conclusion of this article, under the framework of the constrained primer vector, both the impulse addition method and the optimality necessary conditions can be formulated in forms analogous to those in the unconstrained case, thereby revealing the intrinsic connection between constrained and unconstrained problems. Finally, in the typical scenarios analyzed, the optimal flyby-rendezvous trajectory is successfully solved using the constrained cost gradient, and its optimality is verified via the necessary conditions. Meanwhile, a comparison with results obtained via the direct method demonstrates the advantage of the proposed method in terms of optimality and solution efficiency.

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NOMENCLATURE

Notation

t	= Time instant.
X	= State vector of impulse or trajectory.
r	= Position vector.
v	= Velocity vector on impulsive trajectory.
$\tilde{v}$	= Coasting velocity vector on initial orbit, middle target orbit, or final target orbit.
p	= Primer vector.
$\tilde{p}$	= Constrained primer vector.
J	= Impulse cost.
g	= Cost gradient.
Φ	= 6×6 state transition matrix.
M, B	= Matrix functions defined in (20).
A, C, H	= Matrix functions defined in (56).
λ_r, λ_v	= Position and velocity adjoint variables.
Γ, u	= Magnitude and direction of thrust acceleration.
$0_{m \times n}, I_n$	= $m \times n$ Zero matrix, $n \times n$ identity matrix.

Subscript

I	= Beginning of the mission.
F	= Ending of the mission.
0	= Initial impulse.
f	= Final impulse.
c	= Flyby point.
d	= Flyby impulse.
b	= Other endpoint of the flyby trajectory segment adjacent to the flyby impulse, $b \in \{0, f, mk\}$.
a	= Added free impulse.
mk	= k th intermediate free impulse.
rr, rv	= Top-left and top-right 3×3 block of the state transition matrix, respectively.
vr, vv	= Bottom-left and bottom-right 3×3 block of the state transition matrix, respectively.

Subscript

$+$	= Right after a specific instant.
$-$	= Right before a specific instant.
s	= Sign marker: $+$ (right after) for $t_c < t_d$; $-$ (right before) for $t_c > t_d$.
$\bar{s}$	= Inverse sign marker: $-$ (right before) for $t_c < t_d$; $+$ (right after) for $t_c > t_d$.
R	= Classic impulsive rendezvous problem.
y^x	= Superscript notation for state transition matrices: $\Phi^{yx} = \Phi(t_y, t_x)$, denoting the transition from time t_x to t_y ; $x, y \in \{I, F, 0, f, c, d, b, a, mk\}$.

I. INTRODUCTION

Impulsive trajectory planning for spacecraft flyby-rendezvous constitutes a typical and significant class of orbit optimization problems. Such mission scenarios encompass emergency Earth observation [1], [2], space target inspection [3], [4], [5], and asteroid flybys in deep-space

exploration [6], [7], [8], [9], [10], [11]. In emergency Earth observation missions, the spacecraft is required to pass over the event ground location at a specified altitude and ultimately return to its initial orbit. For space target inspection missions, the spacecraft must sequentially conduct flybys of multiple designated space targets to assess the status of malfunctioning satellites or acquire information on noncooperative targets prior to finally returning to the initial orbit or a space station for refueling [12]. In deep-space exploration missions, orbital designs may incorporate flybys of multiple asteroids before reaching the target planet to maximize scientific value.

In the aforementioned scenarios, the mission spacecraft is required to conduct flybys of one or more middle targets within a specified time before rendezvousing with a final target or reaching a designated orbit. For cases with a fixed number of impulses and specified orbital conditions, this problem can be transformed into solving specific nonlinear equations [4], [5]. For the general case, the problem can be formulated as a fixed-time optimal impulsive rendezvous with equality path constraints, which is solvable via direct or indirect methods. Direct methods can be categorized into two types: the first solves the corresponding nonlinear programming problem (NLP) by directly transcribing the problem using approaches such as sequential convex programming (SCP) [13] or analytical gradient methods [14], [15]; the second parameterizes the impulsive optimization problem and then typically solves it via heuristic methods including genetic algorithms [1], [7], [16], [17], differential evolution algorithms [2], [9], [18], [19], particle swarm optimization [20], and ant colony optimization [21]. In recent years, deep learning [8], [22], [23] and reinforcement learning [24] approaches have also emerged that leverage existing samples or constructed databases to predict optimal solutions or estimate initial values. While direct methods exhibit strong applicability and favorable global optimization performance, they generally yield only approximate solutions and fail to provide theoretical insights into trajectory optimality. In contrast, indirect methods rooted in the calculus of variations and the minimum principle can compute exact solutions through analytical or numerical approaches, often revealing the mathematical structure of the optimal solution and offering optimality conditions.

Since its proposal, the primer vector theory, as an indirect method, has been developed into a powerful tool for solving impulsive trajectory optimization problems. Lawden [25] initially provided the first-order necessary conditions for unconstrained multiple-impulse rendezvous trajectories. Lion and Handelsman [26] further derived the cost gradient and criteria for adding an additional impulse, which are used to improve nonoptimal impulsive rendezvous trajectories. Impulsive trajectory optimization based on the primer vector theory generally consists of two key steps. 1) Directly solving the two-point boundary value problem (TPBVP) to obtain an impulsive trajectory that satisfies the boundary conditions. In the inverse-square gravitational field, this process is equivalent to solving the Lambert problem [26], [27]. For perturbed problems and

restricted three-body problems, however, the lack of simple mathematical approaches necessitates solving the TPBVP via numerical integration and shooting method [28], [29]. 2) The optimality of the trajectory is improved using Lion's method [26] until Lawden's necessary conditions are met.

Indirect methods are generally not convenient for handling complex constraints, which is why there have been relatively few studies on the primer vector method with constrained problems so far. Jezewski and Faust [30] investigated a method for handling inequality constraints by introducing a double-wall penalty function to account for arbitrary inequality constraints with specified bounds. In his NASA technical report on primer vector theory, Jezewski [31], [28] further considered perigee constraints and light constraints and later systematically summarized general methods for dealing with arbitrary perturbations and constraints. Prussing et al. [32] studied the primer vector method for direct ascent interception, where the initial position of the vehicle is constrained to the Earth's surface. Building on the work of Prussing and Wellnitz, Taur et al. [33] investigated the problem of considering perigee/apogee inequality path constraints in the optimization of impulsive rendezvous and interception trajectories.

The existing studies based on the primer vector theory have not involved cases with midcourse flyby constraints. Although Lembeck and Prussing [34] studied the problem of impulsive thrust flyby and low-thrust return rendezvous under linear dynamics, they actually considered the flyby-rendezvous process as two separate stages, failing to achieve overall optimality. Considering the advantages of the primer vector theory over direct methods in terms of optimality and efficiency, this article develops a method based on the primer vector theory to solve the fuel-optimal impulsive flyby-rendezvous problem under two-body dynamics. Through the definition of appropriate parameter descriptions, the flyby equality path constraint can be implicitly incorporated into the problem and satisfied by solving the corresponding TPBVPs. By establishing the variational relation of constraint-related parameters, the cost gradient and optimal necessary conditions under the constraint can be derived on the basis of Lion's work.

The main contributions of this article are summarized as follows.

1) By implicitly handling the flyby constraint, the constrained cost gradient and impulse addition criteria are derived, and the optimal necessary conditions under a flyby constraint are proposed, in which the connection between constrained and unconstrained problems is revealed through the constrained primer vector defined in this article.

2) Based on the constrained cost gradient, the optimal impulsive flyby-rendezvous trajectory is successfully solved, and the correctness and effectiveness of the proposed method are verified through comparative simulations with a genetic algorithm.

The rest of this article is organized as follows. Section II defines the parameter description of the flyby-rendezvous problem and analyzes the corresponding TPBVP. Section III

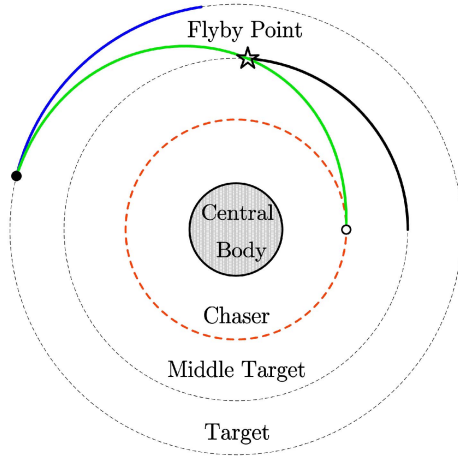


Fig. 1. Diagram of middle target flyby-rendezvous trajectory.

presents the constrained cost gradient with respect to the impulsive trajectory state. Section IV derives the cost variation for impulse addition and defines the constrained primer vector. Section V analyzes the optimal necessary conditions for trajectories under the flyby constraint. Section VI presents and compares the numerical results. Finally, Section VII concludes this article.

II. PROBLEM DESCRIPTION

A. Trajectory State

As shown in Fig. 1, for the two-body fixed-time target flyby-rendezvous problem from t_I to t_F , the chaser is required to first fly by a given middle target before rendezvousing with the final target. It is assumed that the gravitational interaction between the chaser and the middle flyby target is negligible. For cases where gravity cannot be neglected, the velocity change induced by gravitational effects can be treated as a model impulse [13].

Within the scope of this article, only a single middle target is considered. In general, achieving single-target flyby and single-target rendezvous requires three impulses. As shown in Fig. 2, t_c is the flyby time, and t_d is the flyby impulse time. Under the flyby constraint, there are two possible cases. 1) The flyby impulse is applied at the position of the middle target. 2) The middle target lies midway along a two-impulse trajectory segment. The flyby impulse state is defined as $X_d = [t_c \ t_d]^T$; and the initial and final impulse state are defined as $X_0 = t_0$ and $X_f = t_f$, respectively. The flyby-rendezvous impulsive trajectory state in Fig. 2 is defined as

$$X = [X_0 \ X_d^T \ X_f]^T = [t_0 \ t_c \ t_d \ t_f]^T \quad (1)$$

where $t_I \leq t_0 < t_c, t_d < t_f \leq t_F$. When $t_c < t_d$, the middle target lies on the two-impulse trajectory segment between the interval $[t_0, t_d]$, as depicted in Fig. 2(a); when $t_c > t_d$, the middle target lies on the two-impulse trajectory segment between the interval $[t_d, t_f]$, as depicted in Fig. 2(b); when $t_c = t_d$, the flyby impulse is applied at the flyby point.

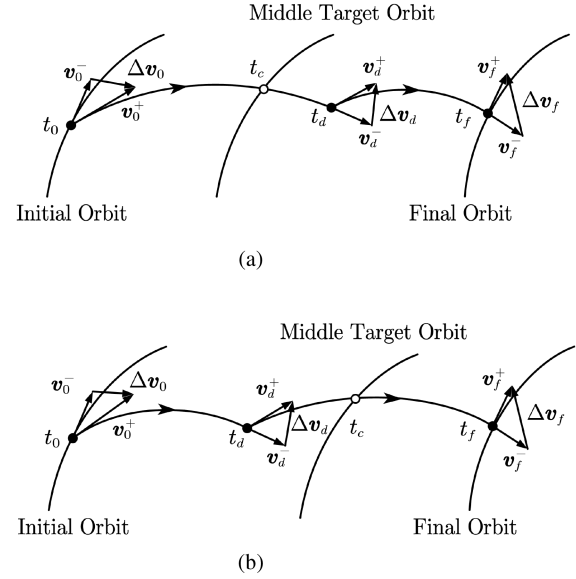


Fig. 2. Flyby-rendezvous problem. (a) $t_c < t_d$, (b) $t_c > t_d$.

Generally, it may be necessary to add additional free impulses to achieve optimality. The state of an intermediate free impulse is defined as $X_{mk} = [r_{mk}^T \ t_{mk}]^T$ ($k = 1, \dots, N_m$, where N_m is the number of free impulses), in which r_{mk} and t_{mk} denote the inertial position and time at which the k th free impulse is applied, respectively [27]. As implied by the term, free impulses feature unconstrained impulse state variables with complete degrees of freedom, allowing their application at arbitrary position and time. Conversely, initial/terminal impulses are restricted to the initial/terminal orbits, whereas flyby impulses are constrained by specified flyby points. The state of a general flyby-rendezvous impulsive trajectory with only one middle target can be expressed as

$$X = [X_0, \ X_{m1}^T, \ \dots, \ X_{mk_c}^T, \ X_d^T, \ X_{m,k_c+1}^T, \ \dots, \ X_{mN_m}^T, \ X_f]^T \quad (2)$$

where the time states of each impulse satisfy their sequential order in X , and $t_{mk_c} < t_c, t_d < t_{m,k_c+1}$.

B. Classical Impulsive Rendezvous

The approach presented in this study is derived from the classical optimal impulsive rendezvous method based on primer vector theory. The two-body impulsive rendezvous problem can be formulated as follows [35]:

$$\begin{aligned} \min J &= \int_{t_I}^{t_F} \Gamma(t) dt = \sum_{k=1}^N \|\Delta v_k\| \\ \text{st. } \dot{\mathbf{r}} &= \mathbf{v}, \quad \dot{\mathbf{v}} = -\mu \mathbf{r}/r^3 + \Gamma \mathbf{u} \end{aligned} \quad (3)$$

where N is the number of impulses, Γ is the thrust magnitude, $\mathbf{u}$ is the thrust direction, and μ is the gravitational constant. Then the Hamiltonian function can be constructed as

$$H = \Gamma + \lambda_r^T \mathbf{v} + \lambda_v^T [-\mu \mathbf{r}/r^3 + \Gamma \mathbf{u}]. \quad (4)$$

By examining (4) and given that $\Gamma \geq 0$, Pontryagin's minimum principle dictates that H attains its minimum value when $\mathbf{u}(t)$ and $\lambda_v(t)$ are collinear and antiparallel at any instant. Owing to the significance of $-\lambda_v(t)$, Lawden [25] coined the term "primer vector" and denoted it as $\mathbf{p}(t)$

$$\mathbf{p}(t) = -\lambda_v(t). \quad (5)$$

The optimal necessary conditions for impulsive trajectories given by Lawden are as follows.

LEMMA 1 Lawden's necessary conditions

- 1) The primer vector $\mathbf{p}$ and its first derivative $\dot{\mathbf{p}}$ are continuous everywhere.
- 2) Impulses occur when $p = 1$, and satisfies $\mathbf{p} = \Delta \mathbf{v} / \|\Delta \mathbf{v}\|$ at impulse times.
- 3) For all $t \in [t_l, t_f]$, $p \leq 1$.
- 4) As a consequence of the three conditions, $dp/dt = \dot{\mathbf{p}} = \dot{\mathbf{p}}^T \mathbf{p} = 0$ at intermediate impulses.

Lawden's necessary conditions can only be used to verify the optimality of a given trajectory. To further solve for the optimal impulsive trajectory, Lion and Handelsman [26] first defined the decision parameters of an impulsive trajectory and solved corresponding TPBVPs to obtain a feasible impulsive trajectory. The cost gradient and impulse addition criterion were then derived to refine nonoptimal impulsive trajectories, yielding optimal necessary conditions equivalent to those of Lawden. The state of a multi-impulse rendezvous trajectory can be defined as

$$\mathbf{X}^R = [X_0 \ X_{m1}^T \ \cdots \ X_{mk}^T \ \cdots \ X_{mNm}^T \ X_f]^T \quad (6)$$

where X_0 , X_f , and X_{mk} are consistent with the definitions given earlier. For the implicit condition required to derive the cost gradient, Lion defined the primer vector at the impulse as a unit vector pointing in the direction of the impulse

$$\mathbf{p}(t_i) = \frac{\Delta \mathbf{v}_i}{\|\Delta \mathbf{v}_i\|} \quad i \in \{0, f, mk\} \quad (k = 1, \dots, N_m). \quad (7)$$

The primer vector and its derivative satisfy the adjoint system, and further details regarding the primer vector theory can be found in [35]. For the trajectory state $\mathbf{X}^R$, the cost variation given by Lion can be written as

$$\begin{aligned} \delta J = & -\dot{\mathbf{p}}_0^T \Delta \mathbf{v}_0 dt_0 - \dot{\mathbf{p}}_f^T \Delta \mathbf{v}_f dt_f \\ & + \sum_{k=1}^{N_m} [(\dot{\mathbf{p}}_{mk}^{+T} - \dot{\mathbf{p}}_{mk}^{-T}) d\mathbf{r}_{mk} \\ & - (\dot{\mathbf{p}}_{mk}^{+T} \mathbf{v}_{mk}^+ - \dot{\mathbf{p}}_{mk}^{-T} \mathbf{v}_{mk}^-) dt_{mk}]. \end{aligned} \quad (8)$$

C. Two-Point Boundary Value Problem

Given the initial, final, and middle target orbital elements as well as the trajectory state $\mathbf{X}$, the boundary conditions for the TPBVPs related to the initial, terminal, and flyby constraints can be determined. In the inverse-square gravitational field, the TPBVPs are satisfied by solving a set of Lambert problems, whose inputs are the boundary inertial position vectors and the transfer time. For given

boundary inputs, the Lambert problem has two groups of solutions corresponding to different transfer directions, where each group associates with distinct multirevolution solutions [36]. In the most general case, for a given trajectory state $\mathbf{X}$, it is necessary to compare combinations of all possible solutions corresponding to the existing TPBVPs to select the impulsive trajectory with the minimum cost.

For the classical multiple-impulse rendezvous problem with only initial, final impulse and intermediate free impulses [27], its trajectory state directly corresponds to the boundary inputs of the Lambert problem, thus facilitating the solution of TPBVPs. In contrast, the issues to be considered for flyby-rendezvous are more complex. As shown in Fig. 2, when $t_c < t_d$, the boundary conditions for the first and second two-impulse trajectory segments are $(\mathbf{r}_0, \mathbf{r}_c, t_c - t_0)$ and $(\mathbf{r}_d, \mathbf{r}_f, t_f - t_d)$, respectively; when $t_c > t_d$, the boundary conditions for the first and second two-impulse trajectory segments are $(\mathbf{r}_0, \mathbf{r}_d, t_d - t_0)$ and $(\mathbf{r}_c, \mathbf{r}_f, t_f - t_c)$, respectively. Due to the existence of a coast segment between t_c and t_d , the inertial position $\mathbf{r}_d$ when the flyby impulse is applied is obtained through the state at t_c and Keplerian extrapolation. Obviously, there is a sequential order in solving the two adjacent boundary value problems.

III. COST GRADIENT

On the basis of Lion's cost variation, the cost gradient for the flyby impulse state can be obtained through direct analysis of the flyby constraint. According to (8), the cost variation of the flyby impulse applied at time t_d can be initially expressed as

$$\delta J_d = (\dot{\mathbf{p}}_d^{+T} - \dot{\mathbf{p}}_d^{-T}) d\mathbf{r}_d - (\dot{\mathbf{p}}_d^{+T} \mathbf{v}_d^+ - \dot{\mathbf{p}}_d^{-T} \mathbf{v}_d^-) dt_d. \quad (9)$$

The application position of the flyby impulse is constrained by the target flyby constraint. To determine the cost variation with respect to the flyby impulse state parameters, it is necessary to find the variational relation between $d\mathbf{r}_d$ and $d\mathbf{X}_d = [dt_c \ dt_d]^T$. As shown in Fig. 2, according to the definition, $\mathbf{r}_d$ is completely determined by Kepler extrapolation from the state at time t_c [for the case of Fig. 2(b), backward time extrapolation is allowed], which is denoted as

$$\mathbf{r}_d = \text{Kepler}(\mathbf{r}_c, \mathbf{v}_c, t_d - t_c). \quad (10)$$

Thus, $d\mathbf{r}_d$ is written as

$$d\mathbf{r}_d = \frac{\partial \mathbf{r}_d}{\partial \mathbf{r}_c} d\mathbf{r}_c + \frac{\partial \mathbf{r}_d}{\partial \mathbf{v}_c} d\mathbf{v}_c + \frac{\partial \mathbf{r}_d}{\partial \Delta t_{dc}} d\Delta t_{dc} \quad (11)$$

where $\mathbf{r}_c$ is the flyby position, $\mathbf{v}_c$ is the flyby velocity, and $\Delta t_{dc} = t_d - t_c$ is the time from the flyby point to the flyby impulse point. On the other hand, the variational state from t_x to t_y satisfies the state transition equation

$$\begin{bmatrix} \delta \mathbf{r}_y \\ \delta \mathbf{v}_y \end{bmatrix} = \Phi(t_y, t_x) \begin{bmatrix} \delta \mathbf{r}_x \\ \delta \mathbf{v}_x \end{bmatrix} = \begin{bmatrix} \Phi_{rr}^{yx} & \Phi_{rv}^{yx} \\ \Phi_{vr}^{yx} & \Phi_{vv}^{yx} \end{bmatrix} \begin{bmatrix} \delta \mathbf{r}_x \\ \delta \mathbf{v}_x \end{bmatrix}. \quad (12)$$

Battin [37] provided an analytical form of the STM under the two-body dynamics. Based on the definition of partial derivatives, $\partial \mathbf{r}_d / \partial \mathbf{r}_c = \Phi_{rr}^{dc}$, $\partial \mathbf{r}_d / \partial \mathbf{v}_c = \Phi_{rv}^{dc}$, $\partial \mathbf{r}_d / \partial \Delta t_{dc} = \mathbf{v}_d^-$ for $t_c < t_d$, and $\partial \mathbf{r}_d / \partial \Delta t_{dc} = \mathbf{v}_d^+$ for $t_c > t_d$ are directly

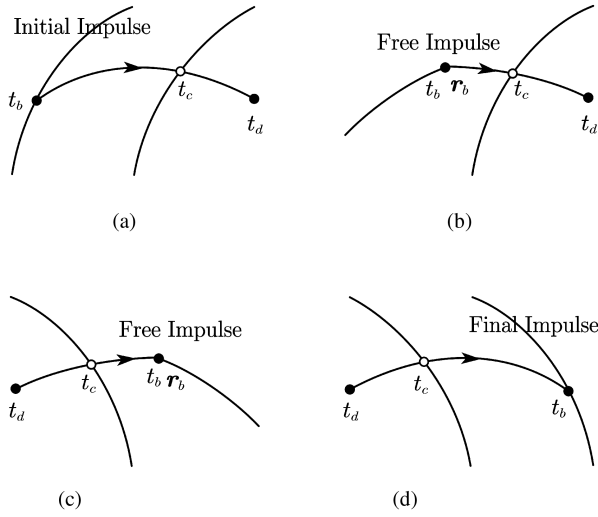


Fig. 3. Four cases of flyby trajectory segment. (a) Case1: $t_b < t_c < t_d$ with $\mathbf{r}_b$ coasting on initial orbit. (b) Case2: $t_b < t_c < t_d$ with $\mathbf{r}_b$ free. (c) Case3: $t_d < t_c < t_b$ with $\mathbf{r}_b$ free. (d) Case4: $t_d < t_c < t_b$ with $\mathbf{r}_b$ coasting on final orbit.

obtained. For the convenience of subsequent expressions, s and $\bar{s}$ are defined to denote the sign markers under different conditions

$$s = \begin{cases} + & t_c < t_d \\ - & t_c > t_d \end{cases}, \quad \bar{s} = \begin{cases} - & t_c < t_d \\ + & t_c > t_d. \end{cases} \quad (13)$$

Then (11) can be expressed as

$$d\mathbf{r}_d = \Phi_{rr}^{dc} d\mathbf{r}_c + \Phi_{rv}^{dc} d\mathbf{v}_c + \mathbf{v}_d^{\bar{s}} dt_d - \mathbf{v}_d^{\bar{s}} dt_c. \quad (14)$$

1) $d\mathbf{r}_c$

For a ground target with a fixed flyby altitude, its position rotates with the Earth's fixed body, so $d\mathbf{r}_c = \mathbf{w}_e \times \mathbf{r}_c dt_c$, where $\mathbf{w}_e$ is the Earth's angular velocity. For the space targets considered in this article, which coast freely in orbit, this yields

$$d\mathbf{r}_c = \tilde{\mathbf{v}}_c dt_c \quad (15)$$

where $\tilde{\mathbf{v}}_c$ is the coasting velocity of the middle target in its own orbit.

2) $d\mathbf{v}_c$

The boundary conditions for the TPBVP corresponding to the flyby trajectory segment (between t_b and t_d) are $(\mathbf{r}_b, \mathbf{r}_c, t_c - t_b)$, where $\mathbf{r}_b$ and t_b are the position and time of the impulse at the other endpoint of the flyby trajectory segment adjacent to the flyby impulse, respectively, with $b \in \{0, f, mk\}$. Thus, $d\mathbf{v}_c$ can be written as

$$d\mathbf{v}_c = \frac{\partial \mathbf{v}_c}{\partial \mathbf{r}_c} d\mathbf{r}_c + \frac{\partial \mathbf{v}_c}{\partial t_c} dt_c + \frac{\partial \mathbf{v}_c}{\partial \mathbf{r}_b} d\mathbf{r}_b + \frac{\partial \mathbf{v}_c}{\partial t_b} dt_b. \quad (16)$$

In analyzing $d\mathbf{v}_c$, four cases need to be considered. 1) The impulse at t_b is the initial boundary impulse, as shown in Fig. 3(a), whose position is constrained by the initial orbit with $d\mathbf{r}_b = \tilde{\mathbf{v}}_0 dt_0$, where $\tilde{\mathbf{v}}_0$ is the coasting velocity of the chaser's initial orbit at t_0 . 2) The impulse at t_b is a free impulse and $t_b < t_c < t_d$, as shown in Fig. 3(b), where the impulse application position $\mathbf{r}_b$ is independent of t_b . 3) The

impulse at t_b is a free impulse and $t_d < t_c < t_b$, as shown in Fig. 3(c), where the impulse application position $\mathbf{r}_b$ is independent of t_b . 4) The impulse at t_b is the terminal boundary impulse, as shown in Fig. 3(d), whose position is constrained by the final target orbit with $d\mathbf{r}_b = \tilde{\mathbf{v}}_f dt_f$, where $\tilde{\mathbf{v}}_f$ is the coasting velocity of the final target at time t_f .

The velocity partial derivatives under the given boundary conditions $(\mathbf{r}_b, \mathbf{r}_c, t_c - t_b)$ are provided in [38] and [39], while Ocampo and Munoz[40] more generally discussed the variational relationships between states and the corresponding partial derivatives in impulsive systems. The derivation of the velocity partial derivatives is given in Appendix A, and according to the results therein, the expressions for the velocity partial derivatives in (16) are obtained as

$$\begin{aligned} \frac{\partial \mathbf{v}_c}{\partial \mathbf{r}_c} &= -\Phi_{rv}^{bc-1} \Phi_{rr}^{bc}, & \frac{\partial \mathbf{v}_c}{\partial \mathbf{r}_b} &= \Phi_{rv}^{bc-1} \\ \frac{\partial \mathbf{v}_c}{\partial t_c} &= \Phi_{rv}^{bc-1} \mathbf{v}_b, & \frac{\partial \mathbf{v}_c}{\partial t_b} &= -\Phi_{rv}^{bc-1} \mathbf{v}_b. \end{aligned} \quad (17)$$

(a) Case 1

In this case, $t_b = t_0 < t_c < t_d$, $\mathbf{v}_b = \mathbf{v}_0^+$, $d\mathbf{r}_b = \tilde{\mathbf{v}}_0 dt_0$, and considering the relationship in (15) and (16) is written as

$$\begin{aligned} d\mathbf{v}_c &= \frac{\partial \mathbf{v}_c}{\partial \mathbf{r}_c} d\mathbf{r}_c + \frac{\partial \mathbf{v}_c}{\partial t_c} dt_c + \frac{\partial \mathbf{v}_c}{\partial \mathbf{r}_b} d\mathbf{r}_b + \frac{\partial \mathbf{v}_c}{\partial t_b} dt_b \\ &= (-\Phi_{rv}^{0c-1} \Phi_{rr}^{0c} \tilde{\mathbf{v}}_c + \Phi_{rv}^{0c-1} \mathbf{v}_0^+) dt_c \\ &\quad + (\Phi_{rv}^{0c-1} \tilde{\mathbf{v}}_0 - \Phi_{rv}^{0c-1} \mathbf{v}_0^+) dt_0. \end{aligned} \quad (18)$$

Substituting (15) and (18) into (14), noting that $\mathbf{v}_d^{\bar{s}} = \mathbf{v}_d^-$ and $\Delta \mathbf{v}_0 = \mathbf{v}_0^+ - \tilde{\mathbf{v}}_0$ in this case, rearranging the expression gives

$$\begin{aligned} d\mathbf{r}_d &= [\mathbf{M}(t_0) \tilde{\mathbf{v}}_c + \mathbf{B}(t_0) \mathbf{v}_0^+ - \mathbf{v}_d^-] dt_c \\ &\quad - \mathbf{B}(t_0) \Delta \mathbf{v}_0 dt_0 + \mathbf{v}_d^- dt_d \end{aligned} \quad (19)$$

where the matrix functions $\mathbf{M}$ and $\mathbf{B}$ are defined, respectively, as

$$\begin{aligned} \mathbf{M}(t_x) &= \Phi_{rr}^{dc} - \Phi_{rv}^{dc} \Phi_{rv}^{xc-1} \Phi_{rr}^{xc} = \Phi_{rv}^{dx} \Phi_{rv}^{cx-1} \\ \mathbf{B}(t_x) &= \Phi_{rv}^{dc} \Phi_{rv}^{xc-1} \end{aligned} \quad (20)$$

where the second equality of $\mathbf{M}$ is derived in Appendix A. Further substituting (19) into (9), this finally yields

$$\begin{aligned} \delta J_d &= (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T [\mathbf{M}(t_0) \tilde{\mathbf{v}}_c + \mathbf{B}(t_0) \mathbf{v}_0^+ - \mathbf{v}_d^-] dt_c \\ &\quad - \dot{\mathbf{p}}_d^{+T} \Delta \mathbf{v}_0 dt_d - (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \mathbf{B}(t_0) \Delta \mathbf{v}_0 dt_0. \end{aligned} \quad (21)$$

(b) Case 2

In this case, $t_b = t_{mk} < t_c < t_d$, $\mathbf{v}_b = \mathbf{v}_{mk}^+$, and $d\mathbf{r}_b$ is unconstrained, (16) is written as

$$\begin{aligned} d\mathbf{v}_c &= (-\Phi_{rv}^{mk,c-1} \Phi_{rr}^{mk,c} \tilde{\mathbf{v}}_c + \Phi_{rv}^{mk,c-1} \mathbf{v}_{mk}^+) dt_c \\ &\quad + \Phi_{rv}^{mk,c-1} d\mathbf{r}_{mk} - \Phi_{rv}^{mk,c-1} \mathbf{v}_{mk}^+ dt_{mk}. \end{aligned} \quad (22)$$

Substituting (15) and (22) into (14) and noting that $\mathbf{v}_d^{\bar{s}} = \mathbf{v}_d^-$ in this case, rearranging the expression gives

$$\begin{aligned} d\mathbf{r}_d &= [\mathbf{M}(t_{mk}) \tilde{\mathbf{v}}_c + \mathbf{B}(t_{mk}) \mathbf{v}_{mk}^+ - \mathbf{v}_d^-] dt_c \\ &\quad + \mathbf{B}(t_{mk}) d\mathbf{r}_{mk} - \mathbf{B}(t_{mk}) \mathbf{v}_{mk}^+ dt_{mk} + \mathbf{v}_d^- dt_d. \end{aligned} \quad (23)$$

Further substituting (23) into (9), this finally yields

$$\begin{aligned} \delta J_d = & (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T [\mathbf{M}(t_{mk}) \tilde{\mathbf{v}}_c \\ & + \mathbf{B}(t_{mk}) \mathbf{v}_{mk}^+ - \mathbf{v}_d^-] dt_c \\ & - \dot{\mathbf{p}}_d^{+T} \Delta \mathbf{v}_d dt_d + (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \mathbf{B}(t_{mk}) d\mathbf{r}_{mk} \\ & - (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \mathbf{B}(t_{mk}) \mathbf{v}_{mk}^+ dt_{mk}. \end{aligned} \quad (24)$$

(c) Case 3

In this case, $t_d < t_c < t_b = t_{mk}$, $\mathbf{v}_b = \mathbf{v}_{mk}^-$, and $d\mathbf{r}_b$ is unconstrained; additionally, noting that $\mathbf{v}_d^s = \mathbf{v}_d^+$, via a derivation analogous to that for Case 2, (9) can be transformed into

$$\begin{aligned} \delta J_d = & (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T [\mathbf{M}(t_{mk}) \tilde{\mathbf{v}}_c \\ & + \mathbf{B}(t_{mk}) \mathbf{v}_{mk}^- - \mathbf{v}_d^+] dt_c \\ & - \dot{\mathbf{p}}_d^{+T} \Delta \mathbf{v}_d dt_d + (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \mathbf{B}(t_{mk}) d\mathbf{r}_{mk} \\ & - (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \mathbf{B}(t_{mk}) \mathbf{v}_{mk}^- dt_{mk}. \end{aligned} \quad (25)$$

(d) Case 4

In this case, $t_d < t_c < t_b = t_f$, $\mathbf{v}_b = \mathbf{v}_f^-$, $d\mathbf{r}_b = \tilde{\mathbf{v}}_f dt_f$, and noting that $\mathbf{v}_d^s = \mathbf{v}_d^+$ and $\Delta \mathbf{v}_f = \tilde{\mathbf{v}}_f - \mathbf{v}_f^-$, via a derivation analogous to that for Case 1, (9) can be converted to

$$\begin{aligned} \delta J_d = & (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T [\mathbf{M}(t_f) \tilde{\mathbf{v}}_c + \mathbf{B}(t_f) \mathbf{v}_f^- - \mathbf{v}_d^+] dt_c \\ & - \dot{\mathbf{p}}_d^{+T} \Delta \mathbf{v}_d dt_d + (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \mathbf{B}(t_f) \Delta \mathbf{v}_f dt_f. \end{aligned} \quad (26)$$

As can be observed, the cost variation of the free impulse in (8) is dependent solely on the impulse state itself, whereas the cost variation of the flyby impulse additionally relies on the state parameters of the adjacent impulse on the flyby trajectory segment. Synthesizing (8), (21), (24), (25), and (26), and taking into account all possible cases of the impulse at t_b , the cost gradient for the trajectory state defined by (2) can be expressed as

$$\mathbf{g}(X) = [\mathbf{g}_0, \mathbf{g}_{m1}^T, \dots, \mathbf{g}_{mk}^T, \mathbf{g}_d^T, \mathbf{g}_{m,k_c+1}^T, \dots, \mathbf{g}_{mN_m}^T, \mathbf{g}_f^T]^T \quad (27)$$

where

$$\begin{aligned} \mathbf{g}_0 = & -\dot{\mathbf{p}}_0^T \Delta \mathbf{v}_0 + \hat{\mathbf{g}}_0, \\ \mathbf{g}_{mk} = & \begin{bmatrix} \dot{\mathbf{p}}_{mk}^+ - \dot{\mathbf{p}}_{mk}^- \\ \dot{\mathbf{p}}_{mk}^{+T} \mathbf{v}_{mk}^- - \dot{\mathbf{p}}_{mk}^{+T} \mathbf{v}_{mk}^+ \end{bmatrix} + \hat{\mathbf{g}}_{mk}, \\ & (k = 1, \dots, N_m) \\ \mathbf{g}_d = & \begin{bmatrix} (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T [\mathbf{M}(t_b) \tilde{\mathbf{v}}_c + \mathbf{B}(t_b) \mathbf{v}_b^s - \mathbf{v}_d^s] \\ -\dot{\mathbf{p}}_d^{+T} \Delta \mathbf{v}_d \end{bmatrix}, \\ & (b \in \{0, f, mk\}) \\ \mathbf{g}_f = & -\dot{\mathbf{p}}_f^T \Delta \mathbf{v}_f + \hat{\mathbf{g}}_f \end{aligned} \quad (28)$$

where the items $\hat{\mathbf{g}}_0$, $\hat{\mathbf{g}}_f$ and $\hat{\mathbf{g}}_{mk}$ satisfy the following conditions:

$$\hat{\mathbf{g}}_0 = \begin{cases} -(\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \mathbf{B}(t_0) \Delta \mathbf{v}_0 & \text{if } t_0 = t_b \\ 0 & \text{else} \end{cases} \quad (29)$$

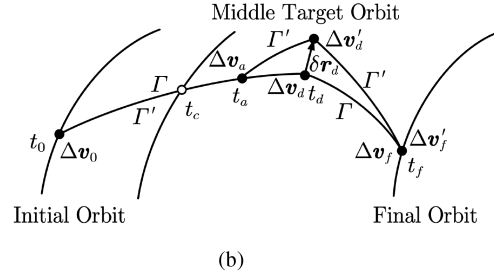
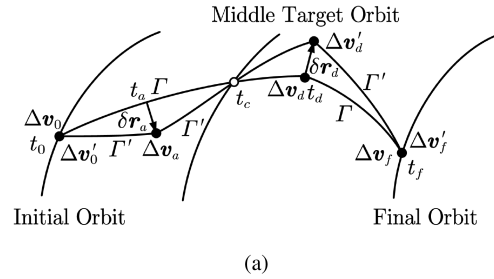


Fig. 4. Impulse addition on flyby trajectory segment. (a) $t_a < t_c$. (b) $t_c < t_a < t_d$.

$$\hat{\mathbf{g}}_f = \begin{cases} (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \mathbf{B}(t_f) \Delta \mathbf{v}_f & \text{if } t_f = t_b \\ 0 & \text{else} \end{cases} \quad (30)$$

$$\hat{\mathbf{g}}_{mk} = \begin{cases} \begin{bmatrix} \mathbf{B}(t_{mk})^T (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-) \\ -(\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \mathbf{B}(t_{mk}) \mathbf{v}_{mk}^s \end{bmatrix} & \text{if } t_{mk} = t_b \\ \mathbf{0}_{4 \times 1} & \text{else.} \end{cases} \quad (31)$$

IV. IMPULSE ADDITION

A. Cost Variation of Impulse Addition

In the flyby-rendezvous impulse trajectory, the method of adding an additional impulse on the unconstrained trajectory segment remains consistent with Lion's approach: Under the first-order approximation, the cost variation after adding the additional impulse can be expressed as [26]

$$\delta J = \|\Delta \mathbf{v}_a\| (1 - \mathbf{p}_a^T \boldsymbol{\eta}_a) \quad (32)$$

where $\Delta \mathbf{v}_a$ is the added extra impulse, and $\boldsymbol{\eta} = \Delta \mathbf{v}_a / \|\Delta \mathbf{v}_a\|$ is the unit direction vector of the extra impulse. Equation (32) shows that, under the first-order approximation, adding an impulse in the direction of $\mathbf{p}_a$ at the point where $p_a > 1$ and p_a reaches its maximum can reduce the cost to the greatest extent.

Denote t_a as impulse addition time, and consider the simplest three-impulse flyby-rendezvous trajectory. For $t_c < t_d$, as shown in Figs. 4 and 5, when deriving the cost variation caused by adding an impulse on the flyby trajectory segment, three cases need to be considered separately: $t_b \leq t_a < t_c$, $t_c \leq t_a \leq t_d$ and $t_a < t_0 = t_b$.

1) $t_a < t_c$: In an impulsive trajectory, the addition of an extra impulse is achieved by introducing a position offset to the reference trajectory at a given time. As illustrated in Fig. 4(a), the costs before and after adding a free impulse

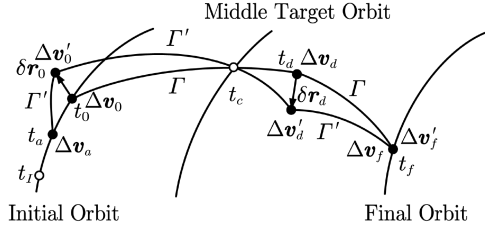


Fig. 5. External impulse addition adjacent to flyby trajectory segment.

during the time interval $t_0 \leq t_a < t_c$ are

$$\begin{aligned} J &= \|\Delta \mathbf{v}_0\| + \|\Delta \mathbf{v}_d\| + \|\Delta \mathbf{v}_f\| \\ J' &= \|\Delta \mathbf{v}_0'\| + \|\Delta \mathbf{v}_a\| + \|\Delta \mathbf{v}_d'\| + \|\Delta \mathbf{v}_f'\|. \end{aligned} \quad (33)$$

Under the first-order approximation, the cost variation is expressed as

$$\begin{aligned} J' - J = \delta J &= \mathbf{p}_0 (\delta \mathbf{v}_0^+ - \delta \mathbf{v}_0^-) + \|\Delta \mathbf{v}_a\| \\ &+ \mathbf{p}_d (\delta \mathbf{v}_d^+ - \delta \mathbf{v}_d^-) + \mathbf{p}_f (\delta \mathbf{v}_f^+ - \delta \mathbf{v}_f^-) \end{aligned} \quad (34)$$

where the definition (7) is utilized.

On each two-impulse trajectory segment, the primer vector and variational state satisfy adjoint relation

$$\mathbf{p} \delta \mathbf{v} - \dot{\mathbf{p}} \delta \mathbf{r} = \text{const}. \quad (35)$$

By utilizing the adjoint relation and boundary conditions $\delta \mathbf{v}_0^- = 0$, $\delta \mathbf{v}_f^+ = 0$, $\delta \mathbf{r}_0 = 0$ and $\delta \mathbf{r}_f = 0$, (34) is converted to

$$\begin{aligned} \delta J &= \|\delta \mathbf{v}_a^+ - \delta \mathbf{v}_a^-\| - \mathbf{p}_a^T (\delta \mathbf{v}_a^+ - \delta \mathbf{v}_a^-) \\ &+ (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \delta \mathbf{r}_d. \end{aligned} \quad (36)$$

From (12), for the state transition equations from t_c to t_a and from t_c to t_d , utilizing $\delta \mathbf{r}_c = 0$ (imposed by the target flyby constraint) and the matrix function $\mathbf{B}$ defined in (20), the relation of $\mathbf{r}_d$ and $\mathbf{r}_a$ is obtained as

$$\delta \mathbf{r}_d = \Phi_{rv}^{dc} (\Phi_{rv}^{ac})^{-1} \delta \mathbf{r}_a = \mathbf{B}(t_a) \delta \mathbf{r}_a. \quad (37)$$

Furthermore, for the state transition equations from t_c to t_a and from t_0 to t_a , as shown in Fig. 4(a), the use of boundary conditions $\delta \mathbf{r}_0 = 0$ and $\delta \mathbf{r}_c = 0$ enables the establishment of

$$\delta \mathbf{v}_a^+ - \delta \mathbf{v}_a^- = \mathbf{A}(t_a) \delta \mathbf{r}_a \quad (38)$$

where $\mathbf{A}(t_a)$ is defined as

$$\mathbf{A}(t_a) = \Phi_{vv}^{ac} (\Phi_{rv}^{ac})^{-1} - \Phi_{vv}^{a0} (\Phi_{rv}^{a0})^{-1}. \quad (39)$$

Substituting (37) and (38) into (36), yields

$$\begin{aligned} \delta J &= \|\Delta \mathbf{v}_a\| - \mathbf{p}_a^T \Delta \mathbf{v}_a \\ &+ (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \mathbf{B}(t_a) \mathbf{A}(t_a)^{-1} \Delta \mathbf{v}_a \\ &= \|\Delta \mathbf{v}_a\| (1 - \tilde{\mathbf{p}}_a^T \boldsymbol{\eta}_a) \end{aligned} \quad (40)$$

where $\tilde{\mathbf{p}}_a$ is defined as

$$\tilde{\mathbf{p}}_a = \mathbf{p}_a - [\mathbf{B}(t_a) \mathbf{A}(t_a)^{-1}]^T (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-). \quad (41)$$

2) $t_c \leq t_a \leq t_d$: Based on the definition of the flyby impulse state $\mathbf{X}_c$, an impulse cannot be directly added on $[t_c, t_d]$. Under the requirement of adding only one impulse, the equivalent way is as shown in Fig. 4(b): replace t_d with t_a as a new flyby impulse time, apply a position variation $\delta \mathbf{r}_d$ at the original flyby impulse time t_d and convert it into a free impulse. The costs before and after the impulse addition are expressed as

$$\begin{aligned} J &= \|\Delta \mathbf{v}_d\| + \|\Delta \mathbf{v}_f\| \\ J' &= \|\Delta \mathbf{v}_a\| + \|\Delta \mathbf{v}_d'\| + \|\Delta \mathbf{v}_f'\|. \end{aligned} \quad (42)$$

Under the first-order approximation, the cost variation is expressed as

$$\begin{aligned} J' - J = \delta J &= \|\Delta \mathbf{v}_a\| + \mathbf{p}_d (\delta \mathbf{v}_d^+ - \delta \mathbf{v}_d^-) \\ &+ \mathbf{p}_f (\delta \mathbf{v}_f^+ - \delta \mathbf{v}_f^-). \end{aligned} \quad (43)$$

By utilizing the adjoint relation and boundary conditions $\delta \mathbf{r}_a = 0$, $\delta \mathbf{v}_f^+ = 0$ and $\delta \mathbf{r}_f = 0$, (43) is converted to

$$\delta J = \|\Delta \mathbf{v}_a\| - \mathbf{p}_a \delta \mathbf{v}_a^+ + (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \delta \mathbf{r}_d. \quad (44)$$

On the other hand, based on the state transition (12) from t_a to t_d and noting that $\delta \mathbf{r}_a = 0$, the following relation is derived:

$$\delta \mathbf{r}_d = \Phi_{rv}^{da} \delta \mathbf{v}_a^+. \quad (45)$$

Substituting (45) into (44) and considering that $\Delta \mathbf{v}_a = \delta \mathbf{v}_a^+$, this gives

$$\begin{aligned} \delta J &= \|\Delta \mathbf{v}_a\| - \mathbf{p}_a^T \Delta \mathbf{v}_a + (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \Phi_{rv}^{da} \Delta \mathbf{v}_a \\ &= \|\Delta \mathbf{v}_a\| (1 - \tilde{\mathbf{p}}_a^T \boldsymbol{\eta}_a) \end{aligned} \quad (46)$$

where

$$\tilde{\mathbf{p}}_a = \mathbf{p}_a - (\Phi_{rv}^{da})^T (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-). \quad (47)$$

In addition, replacing the initial impulse or terminal impulse in Fig. 4 with an intermediate free impulse does not alter the above results.

3) $t_a < t_0$: Within the framework of classical primer vector theory, if $t_l < t_0$ and no impulse can be added inside the boundary impulses (i.e., $p(t) \leq 1$ over $[t_0, t_f]$ and $p(t) > 1$ over $[t_l, t_0]$), an impulse must be added outside the boundary impulse (on the trajectory segment from t_l to t_0), which yields a novel rule for external impulse addition [41]. The primer vector and its derivative outside the boundary impulses are defined as an extrapolation from the adjacent two-impulse trajectory segment. For the addition of an external impulse, the initial impulse time should be substituted by t_a , and a position variation $\delta \mathbf{r}_0$ should be applied at the original initial impulse time t_0 to transform this point into a free impulse. For $t_0 \neq t_b$ (i.e., adding an external impulse to an unconstrained trajectory), the approach has been described in [41]; for $t_0 = t_b$, as shown in Fig. 5, the costs before and after impulse addition are expressed as

$$\begin{aligned} J &= \|\Delta \mathbf{v}_0\| + \|\Delta \mathbf{v}_d\| + \|\Delta \mathbf{v}_f\| \\ J' &= \|\Delta \mathbf{v}_a\| + \|\Delta \mathbf{v}_0'\| + \|\Delta \mathbf{v}_d'\| + \|\Delta \mathbf{v}_f'\|. \end{aligned} \quad (48)$$

Under the first-order approximation, the cost variation is expressed as

$$J' - J = \delta J = \|\Delta \mathbf{v}_a\| + \mathbf{p}_0 (\delta \mathbf{v}_0^+ - \delta \mathbf{v}_0^-) + \mathbf{p}_d (\delta \mathbf{v}_d^+ - \delta \mathbf{v}_d^-) + \mathbf{p}_f (\delta \mathbf{v}_f^+ - \delta \mathbf{v}_f^-). \quad (49)$$

By applying the adjoint relations, the boundary conditions $\delta \mathbf{r}_a = 0$, $\delta \mathbf{v}_f^+ = 0$, $\delta \mathbf{r}_f = 0$, and the continuity of the primer vector derivative at the initial impulse ($\dot{\mathbf{p}}_0^+ = \dot{\mathbf{p}}_0^-$), (49) is transformed into

$$\delta J = \|\Delta \mathbf{v}_a\| + (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \delta \mathbf{r}_d - \mathbf{p}_a \delta \mathbf{v}_a^+. \quad (50)$$

By (12), for the state transition equations from t_c to t_d and from t_c to t_0 , and noting that $\delta \mathbf{r}_c = 0$, the following relation is derived:

$$\delta \mathbf{r}_d = \Phi_{rv}^{dc} \Phi_{rv}^{0c-1} \delta \mathbf{r}_0 = \mathbf{B}(t_0) \delta \mathbf{r}_0. \quad (51)$$

On the other hand, using the state transition equation from t_a to t_0 and the boundary condition $\delta \mathbf{r}_a = 0$, this gives

$$\delta \mathbf{r}_0 = \Phi_{rv}^{0a} \delta \mathbf{v}_a^+. \quad (52)$$

Given that $\Delta \mathbf{v}_a = \delta \mathbf{v}_a^+$, substituting (51) and (52) into (50) yields

$$\begin{aligned} \delta J &= \|\Delta \mathbf{v}_a\| + (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \mathbf{B}(t_0) \Phi_{rv}^{0a} \Delta \mathbf{v}_a - \mathbf{p}_a \Delta \mathbf{v}_a \\ &= \|\Delta \mathbf{v}_a\| (1 - \tilde{\mathbf{p}}_a^T \boldsymbol{\eta}_a) \end{aligned} \quad (53)$$

where

$$\tilde{\mathbf{p}}_a = \mathbf{p}_a - [\mathbf{B}(t_0) \Phi_{rv}^{0a}]^T (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-). \quad (54)$$

For the case where $t_c > t_d$, expressions analogous to (40), (46), and (53) can be derived via a symmetric process. In summary, by comparing with (32), the constrained primer vector is defined as

$$\tilde{\mathbf{p}}(t_a) = \begin{cases} \text{if } (t_b \leq t_a < t_c < t_d) \vee (t_d < t_c < t_a \leq t_b) : \\ \mathbf{p}(t_a) - [\mathbf{B}(t_a) \mathbf{A}(t_a)^{-1}]^T (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-) \\ \text{if } (t_c \leq t_a \leq t_d) \vee (t_d \leq t_a \leq t_c) : \\ \mathbf{p}(t_a) - \mathbf{C}(t_a)^T (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-) \\ \text{if } (t_l \leq t_a \leq t_0 = t_b) \vee (t_f = t_b \leq t_a \leq t_F) : \\ \mathbf{p}(t_a) - [\mathbf{B}(t_a) \mathbf{H}(t_a)]^T (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-) \\ \text{else :} \\ \mathbf{p}(t_a) \end{cases} \quad (55)$$

where $b \in \{0, f, mk\}$, and $\mathbf{A}$, $\mathbf{C}$, and $\mathbf{H}$ are defined as

$$\begin{aligned} \mathbf{A}(t_a) &= \begin{cases} \Phi_{vv}^{ac} \Phi_{rv}^{ac-1} - \Phi_{vv}^{ab} \Phi_{rv}^{ab-1} & t_c < t_d \\ \Phi_{vv}^{ab} \Phi_{rv}^{ab-1} - \Phi_{vv}^{ac} \Phi_{rv}^{ac-1} & t_c > t_d \end{cases} \\ \mathbf{C}(t_a) &= \begin{cases} \Phi_{rv}^{da} & t_c < t_d \\ -\Phi_{rv}^{da} & t_c > t_d \end{cases} \end{aligned}$$

$$\mathbf{H}(t_a) = \begin{cases} \Phi_{rv}^{0a} & t_c < t_d \\ -\Phi_{rv}^{fa} & t_c > t_d \end{cases}. \quad (56)$$

So far, the cost variation after adding an additional impulse on the flyby-rendezvous impulsive trajectory can be expressed in a unified form as

$$\delta J = \|\Delta \mathbf{v}_a\| (1 - \tilde{\mathbf{p}}(t_a)^T \boldsymbol{\eta}_a). \quad (57)$$

B. Impulse Addition Method

Equation (57) indicates that an impulse should be added in the direction of $\tilde{\mathbf{p}}$ at the time when $\tilde{p} > 1$ and $\tilde{p}$ reaches its maximum. In this case, the impulsive cost can be reduced to the greatest extent in the first-order sense. The position offset required for adding an impulse on a flyby-rendezvous trajectory and the application time are determined by

$$\delta \mathbf{r}^* = \begin{cases} \text{if } (t_b \leq t^* < t_c < t_d) \parallel (t_d < t_c < t^* \leq t_b) : \\ \varepsilon \mathbf{A}(t^*)^{-1} \tilde{\mathbf{p}}^* / \|\tilde{\mathbf{p}}^*\| \text{ at } t^* \\ \text{if } (t_c \leq t^* \leq t_d) \parallel (t_d \leq t^* \leq t_c) : \\ \varepsilon \mathbf{C}(t^*) \tilde{\mathbf{p}}^* / \|\tilde{\mathbf{p}}^*\| \text{ at } t_d \\ \text{if } (t_l \leq t^* \leq t_0 = t_b) \vee (t_f = t_b \leq t^* \leq t_F) : \\ \varepsilon \mathbf{H}(t^*) \tilde{\mathbf{p}}^* / \|\tilde{\mathbf{p}}^*\| \text{ at } t_b \end{cases} \quad (58)$$

where t^* is the time with maximum $\tilde{p}$, and ε is taken sufficiently small to ensure that $\delta \mathbf{r}$ satisfies the first-order approximation [35]. For the second case of (58), the flyby impulse time is replaced by t^* , that is, $\mathbf{X}_d = [t_c \ t_d]^T \rightarrow [t_c \ t^*]^T$, and the impulse at time t_d is converted to a free impulse, that is, $\mathbf{X}'_m = [(\mathbf{r}_d + \delta \mathbf{r}^*)^T \ t_d]^T$. For the third case of (58), t^* is replaced with the new boundary impulse time, that is, $\mathbf{X}_0 = t_0 \rightarrow t^*$ or $\mathbf{X}_f = t_f \rightarrow t^*$, while the original boundary impulse is converted into a free impulse, that is, $\mathbf{X}'_m = [(\mathbf{r}_0 + \delta \mathbf{r}^*)^T \ t_0]^T$ or $\mathbf{X}'_m = [(\mathbf{r}_f + \delta \mathbf{r}^*)^T \ t_f]^T$.

The impulse addition on a unconstrained trajectory segment is the same as methods of Lion and Handelsman [26] and Zheng et al. [41].

C. Properties of Constrained Primer Vector

On unconstrained trajectory segments, according to (55), $\tilde{\mathbf{p}} = \mathbf{p}$, and thus $\tilde{\mathbf{p}}$ is continuous across unconstrained trajectory segments. On the constrained trajectory segment, $\tilde{\mathbf{p}}$ has the following properties.

1) Continuity of $\tilde{\mathbf{p}}$ at t_c

When $t_c < t_d$, $\mathbf{A}(t_a)$ can be rewritten as

$$\mathbf{A}(t_a) = [\Phi_{vv}^{ac} - \Phi_{vv}^{ab} \Phi_{rv}^{ab-1} \Phi_{rv}^{ac}] \Phi_{rv}^{ac-1} \quad (59)$$

thus

$$\mathbf{B}(t_a) \mathbf{A}(t_a)^{-1} = \Phi_{rv}^{dc} [\Phi_{vv}^{ac} - \Phi_{vv}^{ab} \Phi_{rv}^{ab-1} \Phi_{rv}^{ac}]^{-1}. \quad (60)$$

Since $\Phi_{rv}^{cc} = \mathbf{0}_{3 \times 3}$ and $\Phi_{vv}^{cc} = \mathbf{I}_3$, the relation $\mathbf{B}(t_c) \mathbf{A}(t_c)^{-1} = \Phi_{rv}^{dc}$ holds for $t_a = t_c$. Similarly, $\mathbf{B}(t_c) \mathbf{A}(t_c)^{-1} = -\Phi_{rv}^{dc}$ when $t_c > t_d$ and $t_a = t_c$. A comparison of the first and second cases in (55) thus reveals the continuity of $\tilde{\mathbf{p}}$ at t_c .

2) Continuity of $\tilde{\mathbf{p}}$ at t_b and $\tilde{\mathbf{p}}_b = \mathbf{p}_b$

When $t_c < t_d$, $\mathbf{A}(t_a)$ can be rewritten as

$$\mathbf{A}(t_a) = [\Phi_{vv}^{ac} \Phi_{rv}^{ac-1} \Phi_{rv}^{ab} - \Phi_{vv}^{ab}] \Phi_{rv}^{ab-1}. \quad (61)$$

If $t_0 \neq t_b$, given $\Phi_{rv}^{bb} = \mathbf{0}_{3 \times 3}$ and $\Phi_{vv}^{bb} = \mathbf{I}_3$, for the case of $t_a = t_b$, $\mathbf{A}(t_b)^{-1} = \mathbf{0}_{3 \times 3}$, leading to the result that $\tilde{\mathbf{p}}_b^+ = \tilde{\mathbf{p}}_b^- = \mathbf{p}_b$ by (55). If $t_0 = t_b$, given that $\mathbf{H}(t_0) = \mathbf{0}_3$, the relation $\tilde{\mathbf{p}}_b^+ = \tilde{\mathbf{p}}_b^- = \mathbf{p}_b$ still holds true. Correspondingly, the same result is obtained for the case where $t_c > t_d$.

3) Continuity of $\tilde{\mathbf{p}}$ at t_d and $\tilde{\mathbf{p}}_d = \mathbf{p}_d$

For the cases of $t_c < t_d$ or $t_c > t_d$, if $t_a = t_d$, $\mathbf{C}(t_a) = \mathbf{0}_3$, and thus $\tilde{\mathbf{p}}_d^+ = \tilde{\mathbf{p}}_d^- = \mathbf{p}_d$ follows from (55).

The above properties indicate that $\tilde{\mathbf{p}}$ is continuous at any time.

V. NECESSARY CONDITION UNDER FLYBY CONSTRAINT

THEOREM 1 Necessary conditions in gradient form under flyby constraint

- 1) $\mathbf{g}(\mathbf{X}) \in \mathbb{G}$;
- 2) for all $t \in [t_l, t_f]$, $\tilde{\mathbf{p}} \leq 1$.

In Theorem 1, $\mathbb{G}$ is defined as the set of gradients $\mathbf{g}(\mathbf{X})$ that satisfy the following conditions:

$$\begin{aligned} (a) \mathbf{g}_{mk} &= 0 \quad (k = 1, \dots, N_m) \\ (b) \mathbf{g}_d &= 0 \vee (t_c = t_d \wedge \mathbf{g}_d^T \mathbf{e}_2 = 0) \\ (c) g_0 &= 0 \vee (t_0 = t_l \wedge g_0 > 0) \\ (d) g_f &= 0 \vee (t_f = t_f \wedge g_f < 0) \end{aligned} \quad (62)$$

where $\mathbf{e}_2 = [1 \ 1]^T$. Condition $\mathbf{g}(\mathbf{X}) \in \mathbb{G}$ is called gradient necessary condition.

PROOF For the impulsive trajectory state $\mathbf{X}$ defined in (2), according to (8), (21), (24), (25), (26), and (27), the cost increment caused by total variation of the state can be written as

$$\Delta J_{dX} = J(\mathbf{X} + d\mathbf{X}) - J(\mathbf{X}) = \mathbf{g}^T d\mathbf{X} + o(\|d\mathbf{X}\|). \quad (63)$$

On the other hand, according to (57), the cost increment caused by impulse addition can be written as

$$\Delta J_{Add} = \|\Delta \mathbf{v}_a\| (1 - \tilde{\mathbf{p}}(t_a)^T \boldsymbol{\eta}_a) + o(\|\Delta \mathbf{v}_a\|) \quad (64)$$

where $\Delta \mathbf{v}_a = \delta \mathbf{v}_a^+ - \delta \mathbf{v}_a^-$. Therefore, according to the principle of linear superposition, the total cost increment considering state variation and impulse addition is expressed as

$$\Delta J = \delta J + o(\sigma) \quad (65)$$

where $\sigma = \max(\|d\mathbf{X}\|, \|\Delta \mathbf{v}_a\|)$, and

$$\delta J = \mathbf{g}^T d\mathbf{X} + \|\Delta \mathbf{v}_a\| (1 - \tilde{\mathbf{p}}(t_a)^T \boldsymbol{\eta}_a). \quad (66)$$

In a sufficiently small neighborhood around $\mathbf{X}$ (i.e., when σ is sufficiently small), ΔJ and δJ have the same sign.

For the first condition of Theorem 1, the following subconditions are proven sequentially.

- (a) $\mathbf{g} = 0$ if $t_c \neq t_d$, $t_0 \neq t_l$, and $t_f \neq t_f$

Since the flyby constraint has been implicitly incorporated into the problem, the variation of the impulsive state $\mathbf{X}$ within its neighborhood is unconstrained if $t_c \neq t_d$, $t_0 \neq t_l$, and $t_f \neq t_f$. Based on (27), if $\mathbf{g} \neq \mathbf{0}$ at the minimum point, it is always possible to set $d\mathbf{X} = -\gamma \mathbf{g}$, with γ taken sufficiently small to ensure that

$$\delta J = -\gamma \left(\|\mathbf{g}_d\|^2 + \|g_0\|^2 + \|g_f\|^2 + \sum_{k=1}^{N_m} \|\mathbf{g}_{mk}\|^2 \right) < 0. \quad (67)$$

Thus, J can be further reduced, which leads to a contradiction with the definition of minimum. Therefore, $\mathbf{g} = 0$ for $t_c \neq t_d$, $t_0 \neq t_l$, and $t_f \neq t_f$.

- (b) $t_c = t_d \wedge \mathbf{g}_d^T \mathbf{e}_2 = 0$

Considering $\delta J = \mathbf{g}_d^T d\mathbf{X}_d$, if $t_c = t_d$ (the case where the flyby impulse is applied at the flyby point), δJ is converted to

$$\delta J = g_{dc} dt_c + g_{dd} dt_d = (g_{dc} + g_{dd}) dt_d. \quad (68)$$

Following the same discussion as in (67), the condition $g_{dc} + g_{dd} = 0$ is derived, which can be summarized as $t_c = t_d \wedge \mathbf{g}_d^T \mathbf{e}_2 = 0$.

- (c) $t_0 = t_l \wedge g_0 > 0$ and $t_f = t_f \wedge g_f > 0$

Considering $\delta J = g_0 dt_0$, if $t_0 = t_l$, the fixed-time constraint $t_l \leq t_0 \leq t_f$ requires $dt_0 > 0$. In this case, if $g_0 > 0$, δJ can no longer be further reduced, meaning the impulsive trajectory has achieved the minimum under the fixed-time constraint. Thus, $t_0 = t_l \wedge g_0 > 0$. The condition $t_f = t_f \wedge g_f > 0$ can be proven through a symmetric discussion.

For the second condition of Theorem 1, without loss of generality, when $d\mathbf{X} = \mathbf{0}$, δJ can be expressed as

$$\delta J = \|\Delta \mathbf{v}_a\| (1 - \tilde{\mathbf{p}}(t_a)^T \boldsymbol{\eta}_a) \quad (69)$$

where $\boldsymbol{\eta}_a = \Delta \mathbf{v}_a / \|\Delta \mathbf{v}_a\|$. If there exists $\|\tilde{\mathbf{p}}(t_a)\| > 1$ for an optimal impulsive trajectory, it is always advisable to set $\Delta \mathbf{v}_a = \epsilon \tilde{\mathbf{p}}(t_a) / \|\tilde{\mathbf{p}}(t_a)\|$, where ϵ is chosen to be sufficiently small such that

$$\delta J = \epsilon (1 - \|\tilde{\mathbf{p}}(t_a)\|) < 0. \quad (70)$$

Thus, the cost J can be further reduced, which leads to a contradiction with the definition of minimum. Therefore, the optimality requires that $\|\tilde{\mathbf{p}}(t_a)\| \leq 1$ for all $t_a \in [t_l, t_f]$.

In summary, Theorem 1 is proven. ■

According to (8), the cost gradient of $\mathbf{X}^R$ defined in (6) is expressed as

$$\mathbf{g}^R(\mathbf{X}^R) = [g_0, \mathbf{g}_{m1}^T, \dots, \mathbf{g}_{mk}^T, \dots, \mathbf{g}_{mN_m}^T, g_f]^T \quad (71)$$

where

$$\begin{aligned} g_0 &= -\dot{\mathbf{p}}_0^T \Delta \mathbf{v}_0, \\ g_f &= -\dot{\mathbf{p}}_f^T \Delta \mathbf{v}_f \\ \mathbf{g}_{mk} &= \begin{bmatrix} \dot{\mathbf{p}}_{mk}^+ - \dot{\mathbf{p}}_{mk}^- \\ \dot{\mathbf{p}}_{mk}^{-T} \mathbf{v}_{mk}^- - \dot{\mathbf{p}}_{mk}^{+T} \mathbf{v}_{mk}^+ \end{bmatrix} \\ &\quad (k = 1, \dots, N_m). \end{aligned} \quad (72)$$

Without considering the flyby impulse, the proof of Theorem 1 is fully applicable to the classical impulsive rendezvous problem. Therefore, Lawden's necessary conditions (Lemma 1) can be equivalently transformed into a formulation analogous to that of Theorem 1.

LEMMA 2 Necessary conditions in gradient form

- 1) $\mathbf{g}^R(\mathbf{X}^R) \in \mathbb{G}^R$;
- 2) for all $t \in [t_I, t_F]$, $p \leq 1$.

In Lemma 2, $\mathbb{G}^R$ is defined as the set of gradients $\mathbf{g}^R(\mathbf{X}^R)$ that satisfy the following conditions:

- (a) $\mathbf{g}_{mk} = 0$ ($k = 1, \dots, N_m$)
 - (b) $g_0 = 0 \vee (t_0 = t_I \wedge g_0 > 0)$
 - (c) $g_f = 0 \vee (t_f = t_F \wedge g_f < 0)$.
- (73)

Theorem 1 can also be transformed into a formulation similar to Lawden's necessary conditions:

THEOREM 2 Necessary conditions under flyby constraint

- 1) The constrained primer vector $\tilde{\mathbf{p}}$ is continuous everywhere.
- 2) The first derivative of constrained primer vector $\dot{\tilde{\mathbf{p}}}$ is continuous everywhere except at flyby time t_c .
- 3) Impulses occur when $\tilde{p} = 1$, and satisfy $\tilde{\mathbf{p}} = \Delta \mathbf{v} / \|\Delta \mathbf{v}\|$ at impulse times.
- 4) For all $t \in [t_I, t_F]$, $\tilde{p} \leq 1$.
- 5) At the free impulses and flyby impulse where $t_c \neq t_d$, $d\tilde{\mathbf{p}}/dt = \dot{\tilde{\mathbf{p}}} = \tilde{\mathbf{p}}^T \tilde{\mathbf{p}} = 0$.

Given the complexity of the proof process, detailed proof of Theorem 2 is presented in Appendix B. Theorem 2 provides an intuitive optimality judgment criterion for the constrained primer vector. A comparison between Theorem 2 and Lemma 1 reveals the inherent connection between constrained and unconstrained problems, and indicates that the necessary conditions for constrained impulsive trajectory optimization still retain a structure analogous to those of unconstrained cases.

The (unconstrained) primer vector theory is actually a limiting case of the constrained primer vector theory proposed in this article. Equation (9) can be derived by reversing the derivation process of (28), during which $\hat{\mathbf{g}}_0$, $\hat{\mathbf{g}}_f$, and $\hat{\mathbf{g}}_{mk}$ are naturally eliminated. Removing the flyby constraint can be regarded as a special case where $t_c = t_d$, with $d\mathbf{r}_c$ and $d\mathbf{v}_c$ being unconstrained (i.e., $d\mathbf{r}_d = d\mathbf{r}_c$ is unconstrained). Therefore, based on (9), the necessary condition for the impulse at t_d is converted to

$$\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^- = 0, \quad \dot{\mathbf{p}}_d^{+T} \mathbf{v}_d^+ - \dot{\mathbf{p}}_d^{-T} \mathbf{v}_d^- = 0. \quad (74)$$

Thus, (28) degenerates into (72), i.e., the constrained cost gradient degenerates into the unconstrained cost gradient. Correspondingly, Theorem 1 degenerates into Lemma 2, and Theorem 2 degenerates into Lemma 1. Furthermore, according to (20) and (56), when $t_d = t_c$, $\mathbf{B} = \Phi_{rv}^{dd} \Phi_{rv}^{ad-1} \equiv \mathbf{0}_{3 \times 3}$ and $\mathbf{C} = \pm \Phi_{rv}^{dd} \equiv \mathbf{0}_{3 \times 3}$. Therefore, based on (55), the

constrained primer vector degenerates into the (unconstrained) primer vector in all cases, which verifies the consistency of the constrained primer vector theory proposed in this article.

Remark: Among the existing primer vector-based methods that account for constraints, the problem of impulsive optimization under equality path constraint of flyby has not yet been discussed. Jezewski [30], [31] addressed the impulsive optimization problem under inequality constraints using the penalty function method, proposing a double-wall penalty function to handle periapsis constraints and lighting constraints. However, this is an approximate approach and difficult to directly apply to scenarios involving equality path constraints. Taur et al. [33] derived the cost gradient under periapsis and apoapsis constraints via the Lagrange multiplier. As the most general method for solving constrained problems, the Lagrange multiplier method requires direct handling of constraints and introduces additional variables. Utilizing the characteristics of flyby constraint, this article analytically defines the state parameters and derives the corresponding cost gradient. The proposed method not only avoids introducing additional Lagrange multipliers but also reduces decision variables by eliminating constrained degrees of freedom through explicit constraint analysis. Notably, equality path constraints are implicitly satisfied via the defined TPBVP, thus obviating the need for direct constraint handling.

Existing works have not discussed the impulse addition criteria and the necessary conditions in problems under flyby constraint. This article fills this research gap and directly reveals the connection with the classical unconstrained problem through the analysis of Theorems 1 and 2.

VI. NUMERICAL RESULT

This section presents numerical examples for solving the fixed-time optimal impulsive flyby-rendezvous.

A. Numerical Method

The proposed iteration-adjustment algorithm (IAA) is developed to solve the flyby-rendezvous optimization. The algorithm initiates with an approximately optimal three-impulse initial trajectory obtained via a coarse grid search. Utilizing the cost gradient, a hybrid iterative method combining the finite-difference Newton's method (FDNM) and steepest descent (SD) is employed to iteratively solve for the impulsive trajectory that satisfies the necessary conditions. This iterative process is referred to as the "internal iteration." After each internal iteration, a series of judgments are made based on the trajectory parameters and optimal necessary conditions to determine whether the iteration should terminate or if impulses need to be deleted, moved, or added, which is termed "external adjustment." The flowchart of the IAA is illustrated in Fig. 6.

In the external adjustment, tiny impulses smaller than a specified threshold $e_{\Delta v}$ are first checked. If

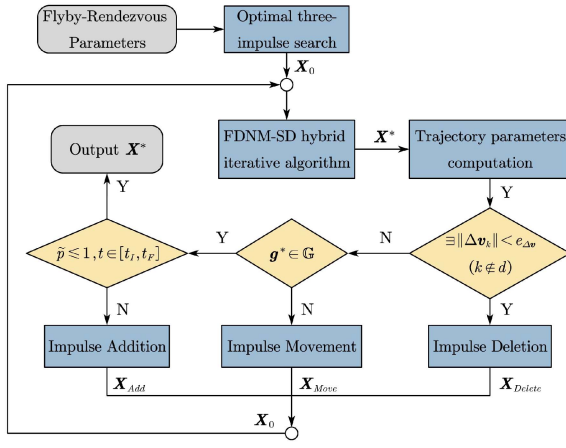


Fig. 6. Iteration-adjustment framework.

such impulses exist, they are removed. In the absence of tiny impulses, the following judgments are made sequentially.

- 1) If $g \in G$ and $p \leq 1$, the necessary conditions are satisfied, and the iteration is terminated.
- 2) If $g \in G$ and $p > 1$, an extra impulse is added.
- 3) If $g \notin G$, it typically indicates the algorithm has encountered iterative stagnation. In this case, the iteration must be restarted by adjusting the trajectory state, i.e., impulse movement.

Impulse movement is heuristic. The strategy adopted in this article is as follows: for each impulse point, find the first adjacent primer vector maximum point (corresponding to time $t_{Adj,k}$) along the direction of increasing primer vector value, select the impulse k^* with the smallest $|t_{Adj,k} - t_k|$ for movement, and replace the original state of this impulse with the state variables corresponding to adding an impulse at t_{Adj,k^*} .

In the internal iteration, the FDNM-SD hybrid iterative method effectively combines the fast convergence property of the Newton method with the cost reduction guarantee of the steepest descent method, avoiding the issues that may arise in the pure Newton method due to the nonpositive definiteness of the Hessian matrix. A simplified implementation of the FDNM-SD hybrid iterative algorithm is presented in Appendix C.

For comparison, this section employs the Broyden-Fletcher-Goldfarb-Shanno (BFGS) quasi-Newton method to solve the given problem within the IAA framework. Its distinction from the FDNM-based IAA lies in the approach to calculating the Hessian matrix. Besides, the genetic algorithm (GA) is also used to directly optimize the trajectory state defined by (2), where the function tolerance is set to $1e-6$. For a given scenario, the GA considers different predefined numbers of impulses and selects the optimal solution from the results of multiple runs.

The simulations was performed in MATLAB 2019a under the Windows 10 system, equipped with an Intel Core i7-8750H CPU @ 2.20 GHz and 32 GB of memory.

TABLE I
Parameter Settings of Departure-Return Flyby-Rendezvous

Parameter	Chaser	Middle target	Final target
R.A.A.N. (deg)	0	0	0
Inclination (deg)	0	0	0
Argument of perigee (deg)	0	0	0
Semi-major axis (km)	10000	11000	10000
Eccentricity (-)	0.01	0.01	0.01
True anomaly (deg)	0	-10	0
Transfer time (-)	$2.0T_p$		

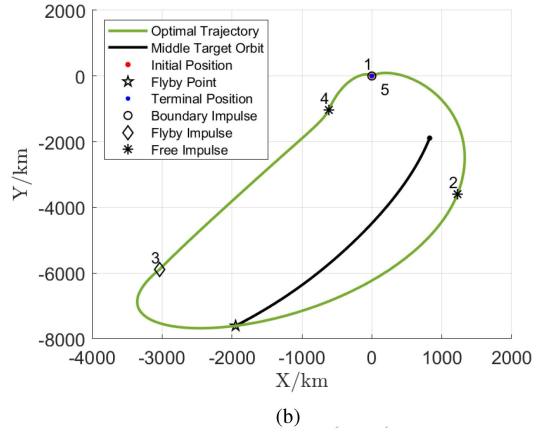
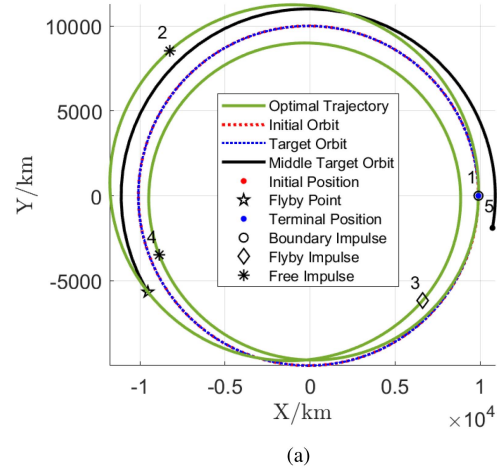


Fig. 7. Trajectories of departure-return flyby-rendezvous. (a) Trajectory in ECI. (b) Trajectory in LVLH of the final target.

B. Departure-Return Flyby-Rendezvous

The “departure-return” is a special case of flyby-rendezvous, where the initial orbital parameters of the chaser and the final target are the same. The parameter settings for this flyby-rendezvous scenario are presented in Table I, where RAAN and T_p denote the right ascension of the ascending node and initial orbital period of the chaser, respectively.

Fig. 7(a) shows the trajectories in Earth-centered inertial (ECI) coordinate system. Here, the red dotted line represents the initial orbit of the chaser, the blue dotted line represents the orbit of the final target, the green solid line represents the flyby-rendezvous trajectory, and the black solid line

TABLE II
Optimal Solution of Departure-Return Flyby-Rendezvous

Impulse type	Impulse (m/s)	Magnitude (m/s)	Time (s)
Initial	[[204.79 255.91 0.00]	327.76	0.00
Free	[93.01 105.01 0.00]	140.28	4179.15
Flyby	[-280.73 -279.83 0.00]	396.38	9885.59
Free	[78.65 -171.84 0.00]	188.98	15710.55
Final	[26.25 92.11 0.00]	95.78	19904.04
Total	1149.17 m/s		

TABLE III
Results Comparison of Departure-Return Flyby-Rendezvous

Method-Impulse number	Optimal impulse (m/s)	Wall-clock time (s)
(Proposed) IAA-FDNM-5	1149.17	16.69
IAA-BFGS-5	1149.17	28.45
GA-3	1249.40	21.39
GA-4	1180.84	163.27
GA-5	1170.00	180.85
GA-6	1176.72	650.24

represents the orbit of the middle target. Fig. 7(b) shows the relative trajectories of the chaser (green solid line) and the middle target (black solid line) in the local vertical local horizontal (LVLH) coordinate system of the final target.

From Fig. 7, the chaser departs from the final target, performs a flyby of the middle target, and ultimately returns to the final target. The overall trajectory forms a spiral tangent to the initial and final orbits. Given that the initial, final, and middle target orbits all lie in the same orbital plane, the optimal flyby-rendezvous trajectory remains confined to the initial orbital plane. The entire mission employs five impulses and includes four coasting segments, with the second segment corresponding to the flyby trajectory segment, with the flyby point between the second and third impulse. The results for the optimal impulsive trajectory are presented in Table II, where the total impulse consumption is 1149.17 m/s.

Table III presents a comparison of IAA-FDNM (the FDNM-based iterative-adjustment algorithm), IAA-BFGS (the BFGS-based iterative-adjustment algorithm), and GA results corresponding to different given numbers of impulses. For GA, multiple runs were performed to avoid poor solutions caused by its inherent random strategy. For 3/4/5/6 impulses, with 4/8/12/16 decision variables, the number of GA solving times is 10/20/30/40, respectively. The initial population size is set to 10 times the number of decision variables, i.e., for 3/4/5/6 impulses, the population sizes are 40/80/120/160, respectively.

From Table III, the solution obtained by IAA-FDNM outperforms the optimal solutions of GA across all impulses, with its cost being 20.82 m/s (1.78%) lower than the optimal cost of GA. The optimal solution of GA corresponds to five impulses, which matches the solution of IAA-FDNM. In terms of solution efficiency, the computation time of IAA-FDNM is 16.69 s, while the computation time of GA's optimal solution (under the optimal impulses) is 180.85 s. The results in Table III demonstrate that GA

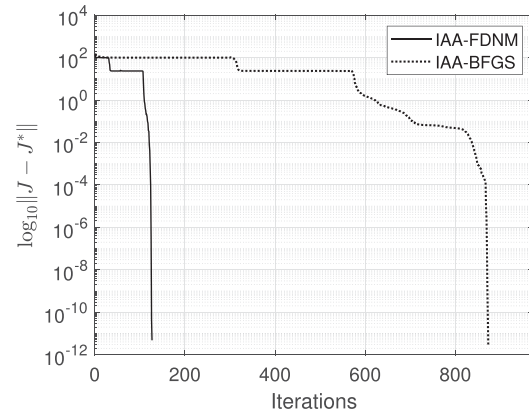


Fig. 8. Convergence curves of departure-return flyby-rendezvous.

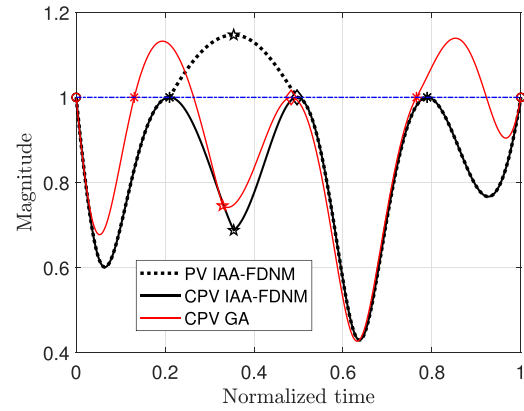


Fig. 9. Constrained primer vector magnitude curves of departure-return flyby-rendezvous.

is significantly inferior to IAA-FDNM in terms of optimality, solution efficiency. Meanwhile, the results of GA also verify the correctness of the optimal solution obtained by IAA-FDNM.

Fig. 8 presents a comparison of the convergence processes of the IAA-FDNM and IAA-BFGS, where IAA-FDNM completes convergence in 127 iterations and IAA-BFGS in 873 iterations. As indicated by Fig. 8 and Table III, FDNM exhibits superior convergence performance and solution efficiency compared with BFGS. While IAA-FDNM undergoes a short plateau phase, it subsequently converges rapidly to the optimal solution in a superlinear manner. In contrast, IAA-BFGS experiences a longer plateau phase followed by a slow decline phase before ultimately converging to the optimal solution. Due to the lower computational cost per iteration of BFGS, the performance gap between IAA-BFGS and IAA-FDNM is far smaller than the ratio of their iterations.

Fig. 9 presents the primer vector magnitude curve (PV IAA-FDNM) and the constrained primer vector magnitude curve (CPV IAA-FDNM) of IAA-FDNM, and the constrained primer vector magnitude curve of GA (CPV GA). The symbol definitions for the impulses are the same as those in Fig. 7. In Fig. 9, the time interval $[0, T_f]$ is linearly mapped to the normalized time interval $[0, 1]$. It can be seen that the constrained primer vector of IAA-FDNM satisfies

TABLE IV
Parameter Settings of Noncoplanar Flyby-Rendezvous

Parameter	Chaser	Middle target	Final target
R.A.A.N. (deg)	0	40	20
Inclination (deg)	0	10	20
Argument of perigee (deg)	0	0	0
Semi-major axis (km)	90000	10000	11000
Eccentricity (-)	0.1	0.1	0.1
True anomaly (deg)	0	40	170
Transfer time (-)	$1.9T_p$		

TABLE V
Optimal Solution of Noncoplanar Flyby-Rendezvous

Impulse type	Impulse (m/s)	Magnitude (m/s)	Time (s)
Initial	[-199.09 -234.19 335.94]	455.34	0.00
Free	[-305.76 -158.79 -362.95]	1405.82	4351.77
Flyby	[-134.74 -58.53 356.36]	385.45	8971.47
Free	[-15.00 -24.33 -239.59]	241.29	12719.80
Final	[104.17 406.94 380.10]	566.50	16144.65
Total	3054.40 m/s		

$\tilde{p}(t) \leq 1$ over $[0, 1]$, and $\dot{\tilde{p}} = 0$ at intermediate impulses. Meanwhile, the constrained primer vector is found to be nonsmooth at the flyby time, i.e., $\dot{\tilde{p}}_c^- \neq \dot{\tilde{p}}_c^+$. Fig. 9 demonstrates that the flyby-rendezvous impulsive trajectory obtained by IAA-FDNM satisfies the necessary conditions under the flyby constraint (Theorem 2). In contrast, the primer vector magnitude curve exceeds 1 across the entire flyby trajectory segment, exhibiting a trend completely opposite to that of the constrained primer vector. This contradiction indicates that the primer vector in the constrained trajectory segment cannot directly provide optimality information.

For comparison, the curve of the constrained primer vector magnitude for GA differs significantly from that of IAA-FDNM and fails to satisfy the condition $\tilde{p}(t) \leq 1$, indicating a nonoptimal solution.

C. Noncoplanar Flyby-Rendezvous

In the noncoplanar flyby-rendezvous, the initial orbits of the chaser, middle target, and final target are all different. The parameter settings for the noncoplanar flyby-rendezvous scenario are given in Table IV.

As shown in Fig. 10, the chaser departs from the initial orbit, flies by the middle target, and then reaches the final target. The entire process uses five impulses and undergoes four coasting segments, among which the second segment is the flyby trajectory segment with the flyby point between the second and third impulse. The optimal solution is presented in Table V, where the total impulse consumption is 3054.40 m/s.

Table VI presents a comparison of results from IAA-FDNM, IAA-BFGS, and GA corresponding to different predefined numbers of impulses. As shown in Table VI, the solution obtained by IAA-FDNM outperforms the optimal solutions of GA across all impulses, with its cost being 27.16 m/s (0.88%) lower than the optimal cost of GA. The optimal solution of GA corresponds to five impulses, which

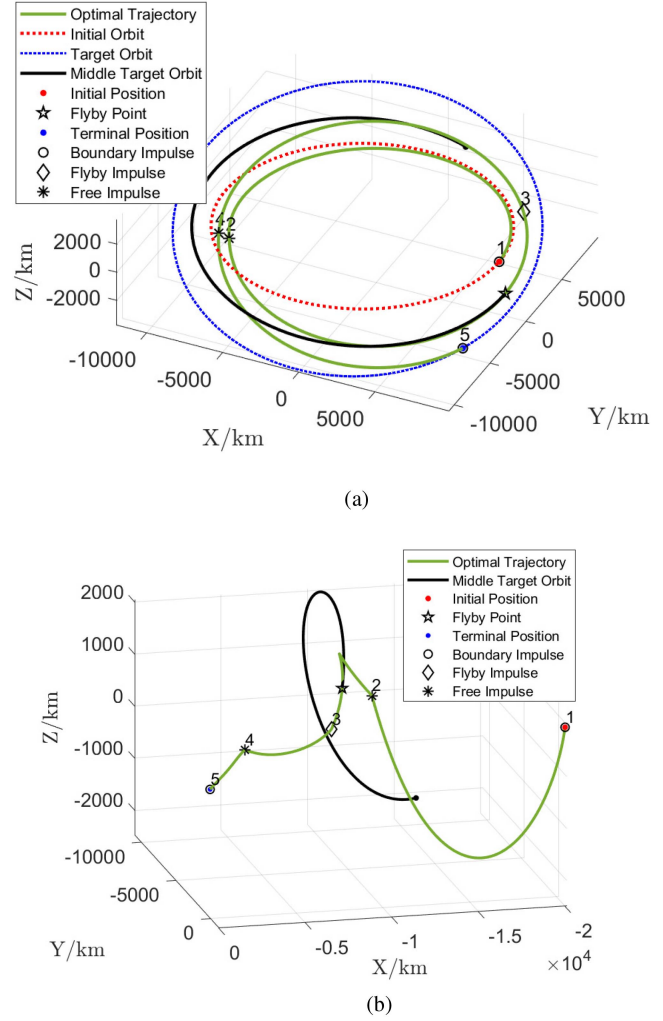


Fig. 10. Trajectories of noncoplanar flyby-rendezvous. (a) Trajectory in ECI. (b) Trajectory in LVLH of the final target.

TABLE VI
Results Comparison of Noncoplanar Flyby-Rendezvous

Method-Impulse number	Optimal impulse (m/s)	Wall-clock time (s)
(Proposed) IAA-FDNM-5	3054.40	21.05
IAA-BFGS-5	3054.40	38.12
GA-3	3585.18	14.32
GA-4	3101.01	87.35
GA-5	3081.57	357.16
GA-6	3098.23	891.68

matches the solution of IAA-FDNM. In terms of solution efficiency, the computation time of IAA-FDNM is 21.05 s, while the computation time of GA's optimal solution (under the optimal impulses) is 357.16 s. These comparison results further demonstrate that GA is inferior to IAA-FDNM in terms of optimality, solution efficiency.

Fig. 11 presents a comparison of the convergence processes of IAA-FDNM and IAA-BFGS, where IAA-FDNM completes convergence in 223 iterations and IAA-BFGS in 1258 iterations. As indicated by Fig. 11 and Table VI, FDNM exhibits superior convergence performance and solution efficiency compared with BFGS. While IAA-FDNM

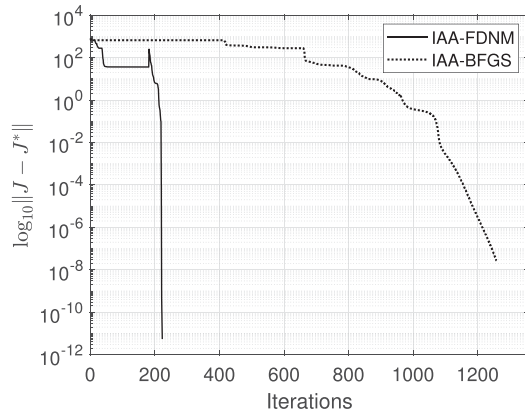


Fig. 11. Convergence curves of noncoplanar flyby-rendezvous.

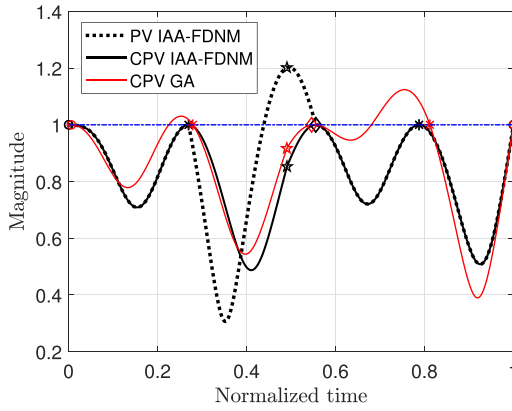


Fig. 12. Constrained primer vector magnitude curves of noncoplanar flyby-rendezvous.

undergoes a short plateau phase and impulse movement adjustment (resulting in a temporary increase in cost), it subsequently converges rapidly to the optimal solution in a superlinear manner. In contrast, IAA-BFGS experiences a longer plateau phase followed by a slow decline phase before ultimately converging to the optimal solution.

From Fig. 12, the optimal constrained primer vector magnitude curve of IAA-FDNM satisfies $\tilde{p}(t) \leq 1$ over the normalized interval $[0, 1]$, and $\dot{\tilde{p}} = 0$ at the intermediate impulses, indicating that the obtained flyby-rendezvous impulsive trajectory satisfies Theorem 2. For comparison, while the impulse distribution of the GA solution is close to that of the optimal solution, the constrained primer vector magnitude curve of GA differs significantly from that of IAA-FDNM, indicating a nonoptimal solution.

VII. CONCLUSION

This article investigates the two-body fuel-optimal impulsive rendezvous problem with a target flyby constraint. The constrained impulsive trajectory state is defined, and the constrained cost gradient is derived, which serves as the foundation for designing an algorithm to solve the optimal impulsive trajectory. The constrained primer vector, discovered during the derivation of cost variation for impulse addition, is proven to be the key for judging the optimality of a constrained impulsive trajectory. This is

reflected in two aspects. 1) The constrained primer vector is the only decision parameter for adding impulses in the constrained trajectory segment. 2) The optimal constrained primer vector needs to satisfy the same conditions as the primer vector at all times except the flyby point. The method of implicitly handling constraints in this article makes up for the deficiency that the indirect method or primer vector theory is difficult to directly analyze constraints. The numerical results demonstrate that the proposed method outperforms the GA in both computation time and fuel consumption.

The work in this article further expands the framework of the primer vector theory and provides new ideas for handling constraints. On the other hand, the proposed necessary conditions under flyby constraint also provide criteria for judging the optimality of results obtained by a direct method. The extension to multiple flybys will be considered in our future research.

APPENDIX

A. VELOCITY PARTIAL DERIVATIVE OF TPBVP

Considering an arbitrary two-impulse trajectory segment from t_x to t_y , the variational states satisfy the state transition equation as (12). The relationships between isochronous variation and nonisochronous variation are expressed as

$$\begin{aligned} d\mathbf{r}_x &= \delta\mathbf{r}_x + \mathbf{v}_x dt_x, & d\mathbf{v}_x &= \delta\mathbf{v}_x + \dot{\mathbf{v}}_x dt_x \\ d\mathbf{r}_y &= \delta\mathbf{r}_y + \mathbf{v}_y dt_y, & d\mathbf{v}_y &= \delta\mathbf{v}_y + \dot{\mathbf{v}}_y dt_y. \end{aligned} \quad (75)$$

Substituting (75) into (12) and rearranging the resulting expression yields

$$\begin{aligned} d\mathbf{v}_x &= -\Phi_{rv}^{yx-1} \Phi_{rr}^{yx} d\mathbf{r}_x + \Phi_{rv}^{yx-1} d\mathbf{r}_y \\ &\quad + [\Phi_{rv}^{yx-1} \Phi_{rr}^{yx} \mathbf{v}_x + \dot{\mathbf{v}}_x] dt_x - \Phi_{rv}^{yx-1} \mathbf{v}_y dt_y \\ d\mathbf{v}_y &= [\Phi_{vr}^{yx} - \Phi_{vv}^{yx} \Phi_{rv}^{yx-1} \Phi_{rr}^{yx}] d\mathbf{r}_x + \Phi_{vv}^{yx} \Phi_{rv}^{yx-1} d\mathbf{r}_y \\ &\quad + [\Phi_{vv}^{yx} \Phi_{rv}^{yx-1} \Phi_{rr}^{yx} - \Phi_{vr}^{yx}] \mathbf{v}_x dt_x \\ &\quad + [\dot{\mathbf{v}}_f - \Phi_{vv}^{yx} \Phi_{rv}^{yx-1} \mathbf{v}_y] dt_y. \end{aligned} \quad (76)$$

Therefore, the following velocity partial derivatives for the boundary value problem $(\mathbf{r}_x, \mathbf{r}_y, t_y - t_x)$ are obtained as:

$$\begin{aligned} \frac{\partial \mathbf{v}_x}{\partial \mathbf{r}_x} &= -\Phi_{rv}^{yx-1} \Phi_{rr}^{yx}, & \frac{\partial \mathbf{v}_x}{\partial \mathbf{r}_y} &= \Phi_{rv}^{yx-1} \\ \frac{\partial \mathbf{v}_y}{\partial \mathbf{r}_x} &= \Phi_{vr}^{yx} - \Phi_{vv}^{yx} \Phi_{rv}^{yx-1} \Phi_{rr}^{yx}, & \frac{\partial \mathbf{v}_y}{\partial \mathbf{r}_y} &= \Phi_{vv}^{yx} \Phi_{rv}^{yx-1} \\ \frac{\partial \mathbf{v}_x}{\partial t_x} &= \dot{\mathbf{v}}_x + \Phi_{rv}^{yx-1} \Phi_{rr}^{yx} \mathbf{v}_x, & \frac{\partial \mathbf{v}_x}{\partial t_y} &= -\Phi_{rv}^{yx-1} \mathbf{v}_y \\ \frac{\partial \mathbf{v}_y}{\partial t_x} &= [\Phi_{vv}^{yx} \Phi_{rv}^{yx-1} \Phi_{rr}^{yx} - \Phi_{vr}^{yx}] \mathbf{v}_x \\ \frac{\partial \mathbf{v}_y}{\partial t_y} &= \dot{\mathbf{v}}_f - \Phi_{vv}^{yx} \Phi_{rv}^{yx-1} \mathbf{v}_y. \end{aligned} \quad (77)$$

Utilizing the symmetry of the state and interchanging the subscripts x and y, the above expressions can also be written

in an equivalent form

$$\begin{aligned}
\frac{\partial \mathbf{v}_x}{\partial \mathbf{r}_x} &= \Phi_{vv}^{xy} \Phi_{rv}^{xy-1}, \quad \frac{\partial \mathbf{v}_x}{\partial \mathbf{r}_y} = \Phi_{vr}^{xy} - \Phi_{vv}^{xy} \Phi_{rv}^{xy-1} \Phi_{rr}^{xy} \\
\frac{\partial \mathbf{v}_y}{\partial \mathbf{r}_x} &= \Phi_{rv}^{xy-1}, \quad \frac{\partial \mathbf{v}_y}{\partial \mathbf{r}_y} = -\Phi_{rv}^{xy-1} \Phi_{rr}^{xy} \\
\frac{\partial \mathbf{v}_x}{\partial t_x} &= \dot{\mathbf{v}}_x - \Phi_{vv}^{xy} \Phi_{rv}^{xy-1} \mathbf{v}_x \\
\frac{\partial \mathbf{v}_x}{\partial t_y} &= [\Phi_{vv}^{xy} \Phi_{rv}^{xy-1} \Phi_{rr}^{xy} - \Phi_{vr}^{xy}] \mathbf{v}_y \\
\frac{\partial \mathbf{v}_y}{\partial t_x} &= -\Phi_{rv}^{xy-1} \mathbf{v}_x, \quad \frac{\partial \mathbf{v}_y}{\partial t_y} = \dot{\mathbf{v}}_y + \Phi_{rv}^{xy-1} \Phi_{rr}^{xy} \mathbf{v}_y. \quad (78)
\end{aligned}$$

From the symmetry of time, the following relationship holds:

$$\begin{aligned}
\frac{\partial \mathbf{v}_x}{\partial t_x} &= -\frac{\partial \mathbf{v}_x}{\partial t_y} = \Phi_{rv}^{yx-1} \mathbf{v}_y \\
\frac{\partial \mathbf{v}_y}{\partial t_y} &= -\frac{\partial \mathbf{v}_y}{\partial t_x} = \Phi_{rv}^{xy-1} \mathbf{v}_x. \quad (79)
\end{aligned}$$

Therefore, when calculating the partial derivatives of velocity respect to time, the following simplest form can be adopted:

$$\begin{aligned}
\frac{\partial \mathbf{v}_x}{\partial t_x} &= \Phi_{rv}^{yx-1} \mathbf{v}_y, \quad \frac{\partial \mathbf{v}_x}{\partial t_y} = -\Phi_{rv}^{yx-1} \mathbf{v}_y \\
\frac{\partial \mathbf{v}_y}{\partial t_x} &= -\Phi_{rv}^{xy-1} \mathbf{v}_x, \quad \frac{\partial \mathbf{v}_y}{\partial t_y} = \Phi_{rv}^{xy-1} \mathbf{v}_x. \quad (80)
\end{aligned}$$

Based on the above results, it can be proven that for any times t_x, t_y, t_z , the following relationship holds:

$$\Phi_{rr}^{zy} - \Phi_{rv}^{zy} \Phi_{rv}^{xy-1} \Phi_{rr}^{xy} = \Phi_{rv}^{zx} \Phi_{rv}^{yx-1}. \quad (81)$$

Since $\Phi^{zy} = \Phi^{zx} \Phi^{xy}$, the block matrices of the STM satisfy

$$\begin{aligned}
\Phi_{rr}^{zy} &= \Phi_{rr}^{zx} \Phi_{rr}^{xy} + \Phi_{rv}^{zx} \Phi_{rv}^{xy} \\
\Phi_{rv}^{zy} &= \Phi_{rr}^{zx} \Phi_{rv}^{xy} + \Phi_{rv}^{zx} \Phi_{vv}^{xy}. \quad (82)
\end{aligned}$$

Thus, substituting the above expression into the left-hand side of (81) yields

$$\begin{aligned}
\Phi_{rr}^{zy} - \Phi_{rv}^{zy} \Phi_{rv}^{xy-1} \Phi_{rr}^{xy} &= (\Phi_{rr}^{zx} \Phi_{rr}^{xy} + \Phi_{rv}^{zx} \Phi_{rv}^{xy}) \\
&\quad - (\Phi_{rr}^{zx} \Phi_{rv}^{xy} + \Phi_{rv}^{zx} \Phi_{vv}^{xy}) \Phi_{rv}^{xy-1} \Phi_{rr}^{xy} \\
&= \Phi_{rv}^{zx} (\Phi_{vr}^{xy} - \Phi_{vv}^{xy} \Phi_{rv}^{xy-1} \Phi_{rr}^{xy}). \quad (83)
\end{aligned}$$

From the equivalence of (77) and (78), the following relationship holds:

$$\frac{\partial \mathbf{v}_x}{\partial \mathbf{r}_y} = \Phi_{vr}^{xy} - \Phi_{vv}^{xy} \Phi_{rv}^{xy-1} \Phi_{rr}^{xy} = \Phi_{rv}^{yx-1}. \quad (84)$$

Finally, substituting the above expression into (83) and (81) is obtained

$$\Phi_{rr}^{zy} - \Phi_{rv}^{zy} \Phi_{rv}^{xy-1} \Phi_{rr}^{xy} = \Phi_{rv}^{zx} \Phi_{rv}^{yx-1}. \quad (85)$$

This relationship is used for the simplification of $\mathbf{M}$ in (20). In addition, since

$$\frac{\partial \mathbf{v}_y}{\partial \mathbf{r}_x} = -\Phi_{rv}^{xy-1} \Phi_{rr}^{xy} = \Phi_{vv}^{yx} \Phi_{rv}^{yx-1}. \quad (86)$$

Equation (81) can also be expressed as

$$\Phi_{rr}^{zy} + \Phi_{rv}^{zy} \Phi_{vv}^{yx} \Phi_{rv}^{yx-1} = \Phi_{rv}^{zx} \Phi_{rv}^{yx-1}. \quad (87)$$

It is worth noting that the above conclusions are independent of the specific dynamics.

B. Equivalence Proof of Theorem 1 and Theorem 2

PROOF Assume that Theorem 1 holds. Under Lion's method, condition 3 of Theorem 2 is always satisfied by (7). The proof of condition 4 in Theorem 2 is identical to that of the second condition in Theorem 1. Since the constrained primer vector and its derivative are continuous at nonimpulsive points, to prove Theorem 2, it is only necessary to demonstrate conditions 1, 2, and 5 at the impulsive points.

1) Unconstrained trajectory segment:

According to (55), $\tilde{\mathbf{p}} = \mathbf{p}$ over the unconstrained trajectory segments, so condition 1 is automatically satisfied.

a) Free impulse:

When Theorem 1 holds, it follows from (28) that for free impulses of unconstrained trajectory segments, the following conditions are satisfied:

$$\begin{aligned}
\dot{\mathbf{p}}_{mk}^+ - \dot{\mathbf{p}}_{mk}^- &= 0 \\
\dot{\mathbf{p}}_{mk}^{-T} \mathbf{v}_{mk}^- - \dot{\mathbf{p}}_{mk}^{+T} \mathbf{v}_{mk}^+ &= 0. \quad (88)
\end{aligned}$$

Since $\dot{\mathbf{p}}_{mk}^+ = \dot{\mathbf{p}}_{mk}^- = \dot{\mathbf{p}}_{mk}$ and $\|\mathbf{p}_{mk}\| = 1$, the following result is further obtained:

$$\begin{aligned}
\dot{\mathbf{p}}_{mk}^{-T} \mathbf{v}_{mk}^- - \dot{\mathbf{p}}_{mk}^{+T} \mathbf{v}_{mk}^+ &= -\dot{\mathbf{p}}_{mk}^T \Delta \mathbf{v}_{mk} \\
&= -\|\Delta \mathbf{v}_{mk}\| \dot{\mathbf{p}}_{mk}^T \mathbf{p}_{mk} = -\|\Delta \mathbf{v}_{mk}\| \dot{\mathbf{p}}_{mk} = 0. \quad (89)
\end{aligned}$$

Thus, conditions 2 and 5 hold at free impulse over unconstrained trajectory segment.

2) Boundary impulse:

Considering the initial impulse, if $t_0 = t_l$, conditions 2 and 5 are automatically satisfied; if $t_0 \neq t_l$, Theorem 1 yields

$$g_0 = -\dot{\mathbf{p}}_0^T \Delta \mathbf{v}_0 = -\|\Delta \mathbf{v}_0\| \dot{\mathbf{p}}_0^T \mathbf{p}_0 = \dot{p}_0 = 0. \quad (90)$$

Thus, condition 5 holds. Furthermore, since the primer vector over $[t_l, t_0]$ is extrapolated from that of the first two-impulse trajectory segment, $\dot{\mathbf{p}}_0^+ = \dot{\mathbf{p}}_0^-$ holds. So condition 2 is automatically satisfied. The same proof process applies to the terminal impulse.

a) Flyby trajectory segment:

The continuity of $\tilde{\mathbf{p}}$ has been verified in Section IV-C, i.e., condition 1 holds. Thus, it is only necessary to prove conditions 2 and 5. Only the case when $t_c < t_d$ is discussed, and the proof for $t_c > t_d$ can be obtained symmetrically.

1) $t_b = t_{mk}$

Considering the case where the impulse at t_b is a free impulse, when Theorem 1 holds, the gradients at both ends of the flyby trajectory segment satisfy the following relations derived from (28):

$$\dot{\mathbf{p}}_b^+ - \dot{\mathbf{p}}_b^- + \mathbf{B}(t_b)^T (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-) = 0 \quad (91)$$

$$\dot{\mathbf{p}}_b^{-T} \mathbf{v}_b^- - \dot{\mathbf{p}}_b^{+T} \mathbf{v}_b^+ - (\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T \mathbf{B}(t_b) \mathbf{v}_b^+ = 0 \quad (92)$$

$$(\dot{\mathbf{p}}_d^+ - \dot{\mathbf{p}}_d^-)^T [\mathbf{M}(t_b) \tilde{\mathbf{v}}_c + \mathbf{B}(t_b) \mathbf{v}_b^+ - \mathbf{v}_d^-] = 0 \quad (93)$$

$$-\dot{\mathbf{p}}_d^{+T} \Delta \mathbf{v}_d = 0. \quad (94)$$

In the subsequent proof, the time derivative of the STM is employed

$$\frac{d}{dt_y} \Phi^{yx} = F \Phi^{yx}, \quad \frac{d}{dt_x} \Phi^{yx} = -\Phi^{yx} F \quad (95)$$

where F denotes the Jacobian matrix of the dynamics: $\dot{r} = v$ and $\dot{v} = f(r, v, t)$, and

$$F = \begin{bmatrix} \mathbf{0}_{3 \times 3} & I_3 \\ F_r & F_v \end{bmatrix}. \quad (96)$$

The following properties are proven when Theorem 1 holds.

$$(a) \dot{\tilde{p}}_b^+ = \dot{\tilde{p}}_b^-$$

For convenience, $A(t_a)$ is expressed in the following form as:

$$A(t_a) = D(t_a) \Phi_{rv}^{ab-1} \quad (97)$$

where $D(t_a) = \Phi_{vv}^{ac} \Phi_{rv}^{ac-1} \Phi_{rv}^{ab} - \Phi_{vv}^{ab}$. By (55), for $t_b \leq t_a < t_c$, taking the time derivative of $\tilde{p}(t_a)$ yields

$$\dot{\tilde{p}} = \dot{p} + D^{-T} [\Phi_{vv}^{ac} \Phi_{rv}^{ac-1} \Phi_{rv}^{ab} + \Phi_{rv}^{ab} D^{-1} \dot{D} - \Phi_{vv}^{ab}]^T B^T (\dot{p}_d^+ - \dot{p}_d^-). \quad (98)$$

When $t_a = t_b$, substituting $\Phi_{rv}^{bb} = \mathbf{0}_3$, $\Phi_{vv}^{bb} = I_3$ and $D(t_b) = -I_3$ into the above equation results in

$$\dot{\tilde{p}}_b^+ = \dot{p}_b^+ + B(t_b)^T (\dot{p}_d^+ - \dot{p}_d^-). \quad (99)$$

Substituting (99) into (91) gives

$$\dot{\tilde{p}}_b^+ = \dot{\tilde{p}}_b^-. \quad (100)$$

Since $\dot{\tilde{p}}_b^-$ lies on the unconstrained trajectory segment, it follows from (55) that $\dot{\tilde{p}}_b^- = \dot{p}_b^-$. Therefore, $\dot{\tilde{p}}$ is continuous at t_b .

$$(b) \dot{\tilde{p}}_d^+ = \dot{\tilde{p}}_d^-$$

When $t_c \leq t_a \leq t_d$, taking the time derivative of $\tilde{p}(t_a)$ and considering $\Phi_{rv}^{da} = -(\Phi_{rr}^{da} + \Phi_{rv}^{da} F_v)$, the following relation is derived:

$$\dot{\tilde{p}} = \dot{p} + (\Phi_{rr}^{da})^T (\dot{p}_d^+ - \dot{p}_d^-). \quad (101)$$

When $t_a = t_d$, substituting $\Phi_{vv}^{dd} = I_3$, $\Phi_{rv}^{dd} = \mathbf{0}_3$, into the above equation yields

$$\dot{\tilde{p}}_d^- = \dot{p}_d^- + (\dot{p}_d^+ - \dot{p}_d^-) = \dot{p}_d^+. \quad (102)$$

Since $\dot{\tilde{p}}_d^+$ lies on the unconstrained trajectory segment, it follows from (55) that $\dot{\tilde{p}}_d^+ = \dot{p}_d^+$. Therefore, $\dot{\tilde{p}}$ is continuous at time t_d . This property does not rely on the validity of Theorem 1.

$$(c) \dot{\tilde{p}}_b^T \tilde{p}_b = 0$$

By (91), (92), and (55), the following result is derived:

$$\begin{aligned} \dot{\tilde{p}}_b^T v_b^- - \dot{\tilde{p}}_b^T v_b^+ + (\dot{p}_b^+ - \dot{p}_b^-)^T v_b^+ &= -\dot{\tilde{p}}_b^T \Delta v_b \\ &= -\|\Delta v_b\| \dot{\tilde{p}}_b^T p_b = -\|\Delta v_b\| \dot{\tilde{p}}_b^T \tilde{p}_b = 0. \end{aligned} \quad (103)$$

Then the continuity of $\dot{\tilde{p}}_b$ implies $\dot{\tilde{p}}_b = \dot{\tilde{p}}_b^T \tilde{p}_b = 0$.

$$(d) \dot{\tilde{p}}_d = \dot{\tilde{p}}_d^T \tilde{p}_d = 0$$

By (94), $\dot{\tilde{p}}_d^T \tilde{p}_d = \dot{p}_d^T \Delta v_d / \|\Delta v_d\| = 0$ is immediately derived, and the continuity of $\dot{\tilde{p}}_d$ implies $\dot{\tilde{p}}_d = \dot{\tilde{p}}_d^T \tilde{p}_d = 0$.

$$(e) \text{Discontinuity of } \dot{\tilde{p}} \text{ at } t_c$$

The proof for general dynamics is relatively complex, so only the case where $F_v = \mathbf{0}_{3 \times 3}$ (e.g., two-body dynamics)

is presented here. For convenience, $A(t_a)$ is rewritten into another form similar to (97) as

$$A(t_a) = E(t_a) \Phi_{rv}^{ac-1} \quad (104)$$

where $E(t_a) = \Phi_{vv}^{ac} - \Phi_{vv}^{ab} \Phi_{rv}^{ab-1} \Phi_{rv}^{ac}$. When $t_b \leq t_a < t_c$, taking the time derivative of $\tilde{p}(t_a)$ gives

$$\dot{\tilde{p}} = \dot{p} + (\Phi_{rv}^{dc} E^{-1} \dot{E} E^{-1})^T (\dot{p}_d^+ - \dot{p}_d^-). \quad (105)$$

Since $E(t_c) = I_3$ and $\dot{E}(t_c) = -\Phi_{vv}^{cb} \Phi_{rv}^{cb-1}$, the above equation is convert to

$$\dot{\tilde{p}}_c^- = \dot{p}_c^- + (\Phi_{rr}^{dc} - \Phi_{rv}^{db} \Phi_{rv}^{cb-1})^T (\dot{p}_d^+ - \dot{p}_d^-) \quad (106)$$

where the relation $\Phi_{rr}^{dc} + \Phi_{rv}^{dc} \Phi_{vv}^{cb} \Phi_{rv}^{cb-1} = \Phi_{rv}^{db} \Phi_{rv}^{cb-1}$ in Appendix A is used. From (101), substituting $t_a = t_c$ gives

$$\dot{\tilde{p}}_c^+ = \dot{p}_c^+ + (\Phi_{rr}^{dc})^T (\dot{p}_d^+ - \dot{p}_d^-). \quad (107)$$

By comparing (106) and (107), it can be concluded that $\dot{\tilde{p}}_c^+ \neq \dot{\tilde{p}}_c^-$, since $\Phi_{rv}^{db} \Phi_{rv}^{cb-1} \neq \mathbf{0}_{3 \times 3}$ in general.

$$2) t_b = t_0$$

If $t_0 = t_l$, conditions 2 and 5 are automatically satisfied. If $t_0 > t_l$, it can be proven that $\dot{\tilde{p}}$ satisfies the following properties.

$$(a) \dot{\tilde{p}}_b^+ = \dot{\tilde{p}}_b^-$$

By (55), for $t_a \leq t_0$, taking the time derivative of $\tilde{p}(t_a)$ yields

$$\begin{aligned} \dot{\tilde{p}} &= \dot{p} - [\dot{B}(t_a) \Phi_{rv}^{0a} \\ &\quad - B(t_a) (\Phi_{rr}^{0a} + \Phi_{rv}^{0a} F_v)]^T (\dot{p}_d^+ - \dot{p}_d^-). \end{aligned} \quad (108)$$

When $t_a = t_0$, substituting $\Phi_{rr}^{00} = I_3$, $\Phi_{rv}^{00} = \mathbf{0}_3$ into the above equation yields

$$\dot{\tilde{p}}_0^- = \dot{p}_0^- + B(t_0)^T (\dot{p}_d^+ - \dot{p}_d^-). \quad (109)$$

Besides, from (91) and $\dot{p}_0^+ = \dot{p}_0^- = \dot{p}_0$, it follows that:

$$B(t_0)^T (\dot{p}_d^+ - \dot{p}_d^-) = 0. \quad (110)$$

By comparing (99) with $t_b = t_0$ and considering $\dot{p}_0^+ = \dot{p}_0^-$, it is derived that $\dot{\tilde{p}}_0^+ = \dot{\tilde{p}}_0^- = \dot{p}_0$. Thus, $\dot{\tilde{p}}_b^+ = \dot{\tilde{p}}_b^-$ for $t_b = t_0$ is proven.

$$(b) \dot{\tilde{p}}_b = \dot{\tilde{p}}_b^T \tilde{p}_b = 0$$

When Theorem 1 holds, the cost gradient of initial impulse derived from (28) satisfies

$$g_0 = -\dot{p}_0^T \Delta v_0 - (\dot{p}_d^+ - \dot{p}_d^-)^T B(t_0) \Delta v_0 = 0. \quad (111)$$

From (110) and $\dot{\tilde{p}}_0^+ = \dot{\tilde{p}}_0^- = \dot{p}_0$, the following relation is derived:

$$\begin{aligned} g_0 &= -\dot{p}_0^T \Delta v_0 = -\|\Delta v_0\| \dot{p}_0^T p_0 \\ &= -\|\Delta v_0\| \dot{\tilde{p}}_0^T \tilde{p}_0 = -\|\Delta v_0\| \dot{\tilde{p}}_0 = 0. \end{aligned} \quad (112)$$

Thus, $\dot{\tilde{p}}_b = \dot{\tilde{p}}_b^T \tilde{p}_b = 0$ for $t_b = t_0$ is proven.

In addition, the validity of $\dot{\tilde{p}}_d^+ = \dot{\tilde{p}}_d^-$, $\dot{\tilde{p}}_d = \dot{\tilde{p}}_d^T \tilde{p}_d = 0$, and discontinuity of $\dot{\tilde{p}}$ at t_c is independent of the impulse type at t_b .

In summary, condition 2 is satisfied, as $\dot{\tilde{p}}_b^+ = \dot{\tilde{p}}_b^-$, $\dot{\tilde{p}}_d^+ = \dot{\tilde{p}}_d^-$, and $\dot{\tilde{p}}_c^+ \neq \dot{\tilde{p}}_c^-$; condition 5 is satisfied, given $\dot{\tilde{p}}_b =$

Algorithm 1: FDNM-SD Hybrid Iteration.

Require: The initial orbital element sets $\mathbf{x}$ of the chaser, middle target, and final target; transfer time T_t ; initial value of trajectory state vector $\mathbf{X}_{(0)}$; state upper boundary $\mathbf{X}_U$; state lower boundary $\mathbf{X}_L$; step computation function $\text{GETSTEP}(\mathbf{x}, T_t, \mathbf{X}, f_{\text{Gdt}})$; cost computation function $\text{GETJ}(\mathbf{x}, T_t, \mathbf{X})$; gradient computation function $\text{GETGDT}(\mathbf{x}, T_t, \mathbf{X})$.

Ensure: Trajectory state after iteration $\mathbf{X}^*$; corresponding gradient $\mathbf{g}^*$.

```
1: Initialize: Let  $k = 1$ ,  $\mathbf{X}_{(k)} = \mathbf{X}_{(0)}$ ,  $f_{\text{Gdt}} = \text{false}$ ,  
    $i_{\text{GD}} = 0$ ; Set iteration control parameters:  
    $i_{\text{GD,max}}, s \in (0, 1)$ ,  $k_{\text{max}}, e_{\Delta X}, e_J$ ;  
2: While  $k \leq k_{\text{max}}$  do  
3:    $J_{(k)} \leftarrow \text{GETJ}(\mathbf{x}, T_t, \mathbf{X}_{(k)})$ ;  $\mathbf{g}_{(k)} \leftarrow \text{GETGDT}(\mathbf{x},$   
      $T_t, \mathbf{X}_{(k)})$ ;  $\Delta \mathbf{X} \leftarrow \text{GETSTEP}(\mathbf{x}, T_t, \mathbf{X}_{(k)}, f_{\text{Gdt}})$ ;  
      $\mathbf{X}_{(k+1)} \leftarrow \mathbf{X}_{(k)} + \Delta \mathbf{X}$ ;  
4:   While there exist elements in  $\mathbf{X}_{(k+1)}$  that do  
     not fall within the upper boundary  $\mathbf{X}_U$  and  
     lower boundary  $\mathbf{X}_L$ , and the impulse times  
     sequence of  $\mathbf{X}_{(k+1)}$  is nonincreasing do  
5:      $\Delta \mathbf{X} \leftarrow s \Delta \mathbf{X}$ ;  $\mathbf{X}_{(k+1)} \leftarrow \mathbf{X}_{(k)} + \Delta \mathbf{X}$ ;  
6:   end while  
7:    $J_{(k+1)} \leftarrow \text{GETJ}(\mathbf{x}, T_t, \mathbf{X}_{(k+1)})$ ;  
8:   While  $J_{(k+1)} > J_{(k)}$  do  
9:     If  $\|\Delta \mathbf{X}\| \leq e_{\Delta X}$  then  $f_{\text{Gdt}} \leftarrow \text{true}$ ; break  
10:     $\Delta \mathbf{X} \leftarrow s \Delta \mathbf{X}$ ;  $\mathbf{X}_{(k+1)} \leftarrow \mathbf{X}_{(k)} + \Delta \mathbf{X}$ ;  
     $J_{(k+1)} \leftarrow \text{GETJ}(\mathbf{x}, T_t, \mathbf{X}_{(k+1)})$ ;  
11:  end while  $\triangleright$  CommentCost reduction  
    judgment; if a reduction is not possible, switch  
    to steepest descent search.  
12:  If  $f_{\text{Gdt}} = \text{true}$  then  
13:     $i_{\text{GD}} \leftarrow i_{\text{GD}} + 1$ ;  
14:    If  $i_{\text{GD}} > i_{\text{GD,max}}$  then  $i_{\text{GD}} \leftarrow 0$ ;  
     $f_{\text{Gdt}} \leftarrow \text{false}$ ;  
15:  end if  $\triangleright$  Record the number of steepest  
    descent iterations, and exit when the  
    maximum value is reached.  
16:  If  $\|J_{(k+1)} - J_{(k)}\| < e_J$  & &  $f_{\text{Gdt}} =$   
     $\text{false}$  then break  $\triangleright$  Satisfy the given cost  
    convergence threshold.  
17:   $k \leftarrow k + 1$ ;  
18: end while  
return:  $\mathbf{X}^* \leftarrow \mathbf{X}_{(k+1)}$ ;  $\mathbf{g}^* \leftarrow \mathbf{g}_{(k+1)}$ 
```

$\dot{\mathbf{p}}_b^T \dot{\mathbf{p}}_b = 0$ and $\dot{\mathbf{p}}_d = \dot{\mathbf{p}}_d^T \dot{\mathbf{p}}_d = 0$. Theorem 2 can be derived when Theorem 1 holds. Conversely, Theorem 1 can be inferred via the reverse process described above when Theorem 2 is satisfied.

Except for the discontinuity of $\dot{\mathbf{p}}$ at t_c , all other properties are independent of the specific dynamics.

C. FDNM-SD Algorithm

The FDNM-SD algorithm is presented in Algorithm 1, where the subscript (k) denotes the k th iteration. All upper bounds of time variables in $\mathbf{X}_U$ are set to t_F , and all lower bounds of time variables in $\mathbf{X}_L$ are set to t_I to ensure the satisfaction of fixed-time constraints. When $f_{\text{Gdt}} = \text{true}$, the function *getStep* directly outputs the negative gradient; when $f_{\text{Gdt}} = \text{false}$, the function *getStep* calculates the Newton step by computing the Hessian matrix via the finite difference method, with the normalized difference step size set to 10^{-8} . As an additional corrective measure, the function *getStep* conducts negative eigenvalue flipping correction on the nonpositive definite Hessian matrix prior to calculating the step size. In the numerical experiments presented in this article, algorithm parameters defined in Algorithm 1 is set as: $i_{\text{Gdt,max}} = 10$, $s = 0.7$, $k_{\text{max}} = 150$, $e_{\Delta X} = 1e - 7$, $e_J = 1e - 11$.

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